

Modular construction of free algebras for distributive combinations of theories

Ohad Kammar¹ and Amaury Mazoyer²

1 Laboratory for Foundations of Computer Science
School of Informatics
University of Edinburgh
Scotland
ohad.kammar@ed.ac.uk

2 Département d'Informatique
ENS de Lyon
France
amaury.mazoyer@ens-lyon.fr

Started: 28 Nov 2022

Abstract

A constructive proof of the construction of free algebras in semi-ring varieties.

1998 ACM Subject Classification F.3.1 Specifying and Verifying and Reasoning about Programs

- §1 and §2 define the 2-categorical and multicategorical notions we need, mostly to fix definitions and notations.
- §3 defines the notion of operad and a 2-functor used in the construction of free algebras (Proposition 3).
- §4 defines the notion of multi-homomorphism and a multi-category and multi-functor used in the construction of free algebras (Proposition 6).
- §5 gives a multi-categorical representability theorem for adjunctions using the Yoneda structure (Proposition 22).
- §6 proves the existence of a left adjoint to a multi-functor under certain conditions (Theorem 23).
- §7 defines the concept of translation and of distributive combination, and proves that the category of models of a distributive combination is a pullback (Proposition 31).
- §8 constructs the free algebra for the distributive combination of two theories and proves that it is indeed free (Lemma 33).

1 2-categories

We'll need basic concepts in higher-dimensional category theory.

A (*strict*) 2-category $\mathcal{C}$ consists of the following data and structure:

- Cellular structure:
 - A collection (set, class, etc.) $\mathcal{C}_0$ of 0-cells $a, b, c, \dots$;
 - A collection (set, class, etc.) $\mathcal{C}_1(a, b)$ of 1-cells $f, g, h, \dots : a \rightarrow b$ for every pair of 0-cells $a, b \in \mathcal{C}_0$;
 - A collection (set, class, etc.) $\mathcal{C}_2(f, g)$ of 2-cells $\alpha, \beta, \gamma, \dots : f \Rightarrow g$ for every pair parallel

of 1-cells $f, g : a \Rightarrow b$. We'll depict 2-cells with oriented faces: $a \begin{array}{c} \xrightarrow{f} \\ \Downarrow \alpha \\ \xrightarrow{g} \end{array} b$.

■ Identities:

- A 1-cell $\text{id}_a : a \rightarrow a$, the *identity* 1-cell, for every 0-cell $a \in \mathbf{C}_0$.
- A 2-cell $\text{id}_f : f \Rightarrow f$, the *identity* 2-cell, for every 1-cell $f : a \rightarrow b$. We depict identities

by marking the face with an equality symbol:
$$a \begin{array}{c} \xrightarrow{f} \\ = \\ \xrightarrow{f} \end{array} b$$

■ Composition:

- A *composition* operation on 1-cells, assigning a 1-cell $g \circ f$ for every pair of composable 1-cells $a \xrightarrow{f} b \xrightarrow{g} c$.
- A *vertical composition* operation on 2-cells, assigning a 2-cell $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $f \xrightarrow{\alpha} g \xrightarrow{\beta} h : a \rightarrow b$, which we depict it by pasting the faces along their shared 1-cell:

$$a \begin{array}{c} \xrightarrow{f} \\ \Downarrow \alpha \\ \xrightarrow{g} \\ \Downarrow \beta \\ \xrightarrow{h} \end{array} b = a \begin{array}{c} \xrightarrow{f} \\ \Downarrow \beta \circ \alpha \\ \xrightarrow{h} \end{array} b$$

- A *horizontal composition* operation on 2-cells, assigning a 2-cell $\beta \bullet \alpha : g_1 \circ f_1 \Rightarrow g_2 \circ f_2$ for every pair of parallel 2-cells $a \xrightarrow{f_1} b \xrightarrow{f_2} c$. We depict it by pasting the faces along the shared vertex:

$$a \begin{array}{c} \xrightarrow{f_1} \\ \Downarrow \alpha \\ \xrightarrow{g_1} \end{array} b \begin{array}{c} \xrightarrow{f_2} \\ \Downarrow \beta \\ \xrightarrow{g_2} \end{array} c = a \begin{array}{c} \xrightarrow{f_2 \circ f_1} \\ \Downarrow \beta \bullet \alpha \\ \xrightarrow{g_2 \circ g_1} \end{array} c$$

such that:

- 1-cell composition is associative and identities are neutral:

$$h \circ (g \circ f) = (h \circ g) \circ f \quad \text{id}_b \circ f = f = f \circ \text{id}_a$$

- 2-cell vertical composition is associative and identities are neutral:

$$\gamma \circ (\beta \circ \alpha) = (\gamma \circ \beta) \circ \alpha \quad \text{id}_g \circ \alpha = \alpha = \alpha \circ \text{id}_f$$

- 2-cell horizontal composition is associative and identities between identities are neutral, and composition of 2-cell identities preserves 1-cell composition:

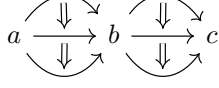
$$\gamma \bullet (\beta \bullet \alpha) = (\gamma \bullet \beta) \bullet \alpha \quad \text{id}_{\text{id}_b} \bullet \alpha = \alpha = \alpha \bullet \text{id}_{\text{id}_a} \quad \text{id}_g \bullet \text{id}_f = \text{id}_{g \circ f}$$

- The interchange law holds:

$$(\beta_2 \circ \alpha_2) \bullet (\beta_1 \circ \alpha_1) = (\beta_2 \bullet \beta_1) \circ (\alpha_2 \circ \alpha_1)$$

$$\begin{array}{c} a \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow \alpha_1 \\ \xrightarrow{\quad} \end{array} b \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow \alpha_2 \\ \xrightarrow{\quad} \end{array} c \\ \circ \end{array} = a \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow \beta_1 \bullet \beta_2 \\ \xrightarrow{\quad} \end{array} b \bullet b \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow \alpha_1 \circ \alpha_2 \\ \xrightarrow{\quad} \end{array} c$$

This law means we can depict compositions of 2-cells using commutative diagrams, since all decompositions of the diagram into pasting expressions are equal 2-cells. For example, we will depict both expressions in the interchange law by this pasting diagram:



► **Example 1.** The paradigmatic example is the 2-category **CAT** that has:

- potentially large categories as 0-cells;
- functors between them as 1-cells;
- natural transformations between them as 2-cells;
- pointwise composition of functors as 1-cell composition;
- pointwise composition of natural transformations as horizontal composition;
- horizontal composition of natural transformations given, for $\mathcal{A} \xrightarrow{F_1} \mathcal{B} \xrightarrow{F_2} \mathcal{C}$ by:

$$(\beta \bullet \alpha)_a = \beta_{G_1 a} \circ F_2 \alpha_a \quad \text{i.e., by the naturality square:} \quad \begin{array}{ccc} F_2 F_1 a & \xrightarrow{\beta_{F_1 a}} & G_2 F_1 a \\ F_2 \alpha_a \downarrow & (\beta \bullet \alpha)_a \searrow & \downarrow G_2 \alpha_a \\ F_2 G_1 a & \xrightarrow{\beta_{G_1 a}} & G_2 G_1 a \end{array}$$

The required properties are well-known (Mac Lane, 1978). ◀

An *adjunction* in a 2-category $\mathcal{C}$ consists of:

- two 1-cells $a \xrightarrow{f} b \xrightarrow{u} a$ called the left (f) and right (u) adjoints;
- two 2-cells $u \circ f \xrightarrow{\eta} \text{id}_a$ and $\text{id}_b \xrightarrow{\varepsilon} u \circ f$ called the unit (η) and counit (ε).

such that:

$$\text{■ } a \xrightarrow{f} b \xrightarrow{u} a \xrightarrow{f} b = \text{id}_f \quad \text{and} \quad b \xrightarrow{u} a \xrightarrow{f} b \xrightarrow{u} a = \text{id}_u$$

► **Example 2.** An adjunction in **CAT** is the ordinary notion of an adjunction. ◀

A 2-*functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of three mappings:

- $F_0 : \mathcal{C}_0 \rightarrow \mathcal{D}_0$ sending 0-cells a in $\mathcal{C}$ to 0-cells $F_0 a$ in $\mathcal{D}$.
- $F_1 : \mathcal{C}_1(a, b) \rightarrow \mathcal{D}_1(F_0 a, F_0 b)$ sending 1-cells $f : a \rightarrow b$ in $\mathcal{C}$ to 1-cells $F_1 f : F_0 a \rightarrow F_0 b$ in $\mathcal{D}$.
- $F_2 : \mathcal{C}_2(f, g) \rightarrow \mathcal{D}_2(F_1 f, F_1 g)$ sending 2-cells $a \xrightarrow{f} b \xrightarrow{g} c$ in $\mathcal{C}$ to 2-cells $F_0 a \xrightarrow{F_1 f} F_0 b \xrightarrow{F_1 g} F_0 c$ in $\mathcal{D}$.

such that F preserves all identities and compositions.

The point of 2-functors is that they transport pasting diagrams to pasting diagrams, while preserving the identities. Our main usage of 2-functors is the following lemma that lets us transport adjunctions:

► **Lemma 1.** Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a 2-functor. If $\langle f : a \rightarrow b, u, \eta, \varepsilon \rangle$ is an adjunction in $\mathcal{C}$ then $\langle Ff : Fa \rightarrow Fb, Fu, F\eta, F\varepsilon \rangle$ is an adjunction in $\mathcal{D}$.

Proof

Type-checking, the candidate adjunction has the right type. Since 2-functors preserve pasting diagrams and identities:

$$Fa \xrightarrow{Ff} Fb \xrightarrow{Fu} Fa \xrightarrow{Ff} Fb = \text{Fid}_f = \text{id}_{Ff} \quad Fb \xrightarrow{Fu} Fa \xrightarrow{Ff} Fb \xrightarrow{Fu} Fa = \text{Fid}_u = \text{id}_{Fu}$$

and so this candidate is in fact an adjunction. $\blacksquare$

We will also observe several (strict) natural 2-transformations. Let $F, G : \mathbf{C} \rightarrow \mathbf{D}$ be two 2-functors between 2-categories. A (strict) natural 2-transformation $\mathbf{C} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \Xi \\ \xrightarrow{G} \end{array} \mathbf{D}$ consists

of:

- A 1-cell $\Xi_a : F_0a \rightarrow G_0a$, for each 0-cell a ;

such that:

- For every 1-cell $f : a \rightarrow b$ in $\mathbf{C}$:

$$\begin{array}{ccc} F_0a & \xrightarrow{F_1f} & F_0b \\ \Xi_a \downarrow & = & \downarrow \Xi_b \\ G_0a & \xrightarrow{G_1f} & G_0b \end{array}$$

- For every 2-cell $a \begin{array}{c} \xrightarrow{f} \\ \Downarrow \alpha \\ \xrightarrow{g} \end{array} b$:

$$\begin{array}{ccc} & Ga & \\ \Xi_a \nearrow & & \searrow Gf \\ Fa & & Gb \\ Fg \searrow & & \nearrow \Xi_b \\ & Fb & \end{array} \quad = \quad \begin{array}{ccc} & Ga & \\ \Xi_a \nearrow & & \searrow Gf \\ Fa & & Gb \\ Fg \searrow & & \nearrow \Xi_b \\ & Fb & \end{array}$$

2 Multicategories

A multicategory $\mathcal{U}$ consists of:

- A collection (set, class, etc.) $\text{Ob}(\mathcal{U})$ of objects $a, b, c, \dots$
- A collection (set, class, etc.) $\mathcal{U}(a_1, \dots, a_n; b)$ of multi-morphisms, for every (possibly empty) list of source objects $a_1, \dots, a_n$ and target object b , in which case we write $f : (a_1, \dots, a_n) \rightarrow b$ and depict this multi-morphisms like so:

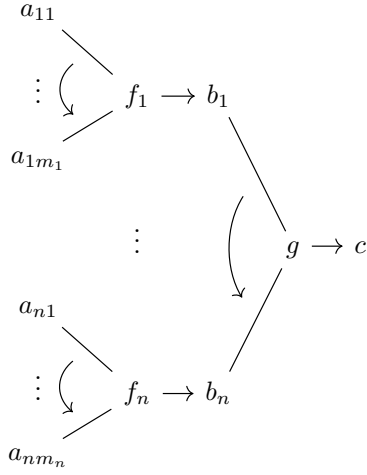
$$\begin{array}{c} a_1 \\ \vdots \\ a_n \end{array} \begin{array}{c} \diagdown \\ \diagup \end{array} f \rightarrow b$$

The *morphisms* are those multi-morphisms whose source is of length 1.

- An *identity* morphism $\text{id}_a : (a) \rightarrow a$ for each object a .

We'll depict morphisms by single-source arrows and identity morphisms by elongated equality symbols, as in ordinary commutative diagrams.

- The *composition* operation, defined for every multi-morphism $g : (b_1, \dots, b_n) \rightarrow c$ and compatible list of multi-morphisms $f_i : (a_{i1}, \dots, a_{im_i}) \rightarrow b_i$, for each $1 \leq i \leq n$, yielding a multi-morphism $g \circ (f_1, \dots, f_n) : (a_{11}, \dots, a_{1m_1}, \dots, a_{n1}, \dots, a_{nm_n}) \rightarrow c$. We depict the composed morphism by pasting the shared objects:



such that:

- identities are neutral, i.e., for all $f : (a_1, \dots, a_n) \rightarrow b$:

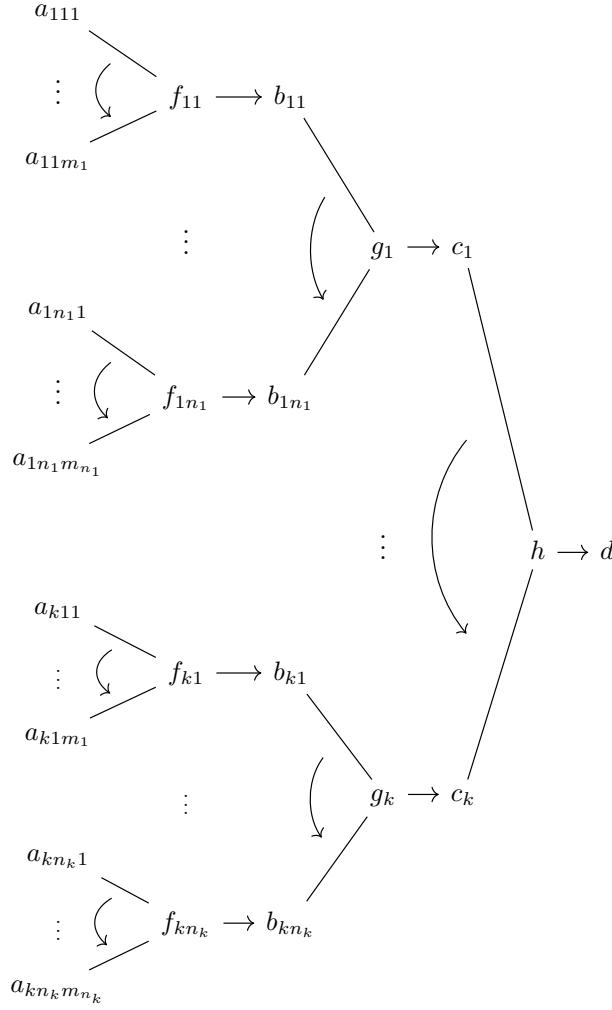
$$f \circ (\text{id}_{a_1}, \dots, \text{id}_{a_n}) = f = \text{id}_b \circ f$$

- composition is associative:

$$\begin{aligned} h \circ (g_1 \circ (f_{11}, \dots, f_{1m_1}), \dots, g_n \circ (f_{n1}, \dots, f_{nm_n})) \\ = (h \circ (g_1, \dots, g_n)) \circ (f_{11}, \dots, f_{1m_1}, \dots, f_{n1}, \dots, f_{nm_n}) \end{aligned}$$

Associativity means that, like for ordinary commutative diagrams, we may decompose a multi-morphism diagram in any several equivalent ways. For example, we depict the

multi morphism of the associativity equation like so:



The paradigmatic example is multi-linear maps:

► **Example 3.** Let $\langle \mathcal{C}, \Pi \rangle$ be a category with specified finite products. We equip it with a multicategory structure as follows:

- The objects are the same, $\text{Ob}(\langle \mathcal{C}, \Pi \rangle) := \text{Ob}(\mathcal{C})$.
- A multi-morphism $f : (a_1, \dots, a_n) \rightarrow b$ is any morphism $f : \prod_{i=1}^n a_i \rightarrow b$.
- The identities rely on the canonical isomorphism:

$$\text{id}_a := \prod_{i=1}^1 a \cong a$$

- Similarly, composition relies on a canonical isomorphism. Letting:
 - $[n] := \{1, \dots, n\}$ be the enumeration of the index set;
 - $p := \sum_{i=1}^n m_i$ the total number of objects in the source of the composition morphism; and
 - $\varphi : [p] \xrightarrow{\cong} \coprod_{i=1}^n [m_i]$ be an enumeration of the nested set of indices that's compatible with the source list of the composition;

define:

$$g \circ (f_1, \dots, f_n) : \prod_{\ell=1}^p a_{\varphi \ell} \xrightarrow{\cong} \prod_{i=1}^n \prod_{j=1}^{m_i} a_{i,j} \xrightarrow{\prod_i f_i} \prod_{i=1}^n b_i \xrightarrow{g} c$$

- Tedious verification of the axioms.

◀

Here is a slightly different way to construct a similar multicategory, but this time without postulating a choice of finite products. Not sure it's a good way to do it.

► **Example 4.** Let $\mathcal{C}$ be a category that merely has finite products.

- Objects are the same.
- A multi-morphism $f : (a_1, \dots, a_n) \rightarrow b$ consists of a chosen product $\text{Dom}(f)$ for $\pi_{i=1}^n a_i$ together with a morphism $f : \text{Dom}(f) \rightarrow b$. For identities id_a we can then take the chosen product $\prod_{i=1}^1 a_i$ to be a itself together with the identity, and for composition we can take $\prod_{i=1}^n \prod_{j=1}^{m_i} a_{i,j}$, although here we need to employ some choice for the outer product.

◀

Generalising the example $\langle \mathcal{C}, \Pi \rangle$:

A *multi-functor* $F : \mathcal{U} \rightarrow \mathcal{V}$ consists of two mappings:

- $F : \text{Ob}(\mathcal{U}) \rightarrow \text{Ob}(\mathcal{V})$ sending $\mathcal{U}$ objects to $\mathcal{V}$ objects; and
- $F : \mathcal{U}(a_1, \dots, a_n; b) \rightarrow \mathcal{V}(Fa_1, \dots, Fa_n; Fb)$ sending a multi-morphism $f : (a_1, \dots, a_n) \rightarrow b$ to a multi-morphism $Ff : (Fa_1, \dots, Fa_n) \rightarrow Fb$.

such that F preserves:

- identities: $F\text{id}_a = \text{id}_{Fa} : (Fa) \rightarrow a$; and
- composition: $F(g \circ (f_1, \dots, f_n)) = Fg \circ (Ff_1, \dots, Ff_n)$.

As usual, multi-functors are defined to preserve the structure of diagrams with multi-morphisms.

► **Example 5.** The identity multi-functor $\text{Id} : \mathcal{U} \rightarrow \mathcal{U}$ sends every object and multi-morphism to themselves. The composition of multi-functors is a multi-functor.

◀

A *multi-natural transformation* $\mathcal{U} \begin{matrix} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{matrix} \mathcal{V}$ between multi-functors consists of:

- a morphism $\alpha_a : Fa \rightarrow Ga$ for every object $a \in \text{Ob}(\mathcal{U})$

such that:

■

$$\begin{array}{ccc}
 Fa_n & \xrightarrow{\alpha_{a_n}} & Ga_n \\
 \vdots & & \vdots \\
 Fa_1 & \xrightarrow{\alpha_{a_1}} & Ga_1 \\
 \swarrow & & \searrow \\
 Ff & & Gf \\
 \downarrow & = & \downarrow \\
 Fb & \xrightarrow{\alpha_b} & Gb
 \end{array}$$

We can define horizontal and vertical composition of multi-natural transformations analogously to their category theoretic way:

$$(\beta \circ \alpha)_a := \beta_a \circ (\alpha_a) \quad (\beta \bullet \alpha)_a := \beta_{G_1 a} \circ (F_2 \alpha_a) = G_2 \alpha_a \circ (\beta_{F_1 a})$$

and they are multi-natural for analogous reasons. As a consequence we have a 2-category **MultiCAT** where:

- 0-cells are multi-categories;
- 1-cells are multi-functors; and
- 2-cells are multi-natural transformations
- with multi-functor composition and identities:

$$\text{Id}_C a := a \quad \text{Id}_C f := f$$

vertical, horizontal composition of multi-natural transformations, with identities:

$$(\text{id}_F)_a := \text{id}_{F a} : (a) \rightarrow a$$

For example, we have a 2-functor structure $-\circ : \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$:

- sending each multicategory $\mathcal{U}$ to its (ordinary) *category of morphisms*:
 - $\text{Ob}(\mathcal{U}_\circ) := \text{Ob}(\mathcal{U})$;
 - $\mathcal{U}_\circ(a, b) := \mathcal{U}(a; b)$, i.e., the morphisms in $\mathcal{U}_\circ$ are the multi-morphisms with a length-1 source.
 - identities and composition are given by identity morphisms and composition in the multi-category.
- sending each multi-functor $F : \mathcal{U} \rightarrow \mathcal{V}$ to the functor:
 - sending $a \in \text{Ob}(\mathcal{U})$ to $F a \in \text{Ob}(\mathcal{V})$;
 - sending $f : (a) \rightarrow b$ to $F_\circ f := F f : (F a) \rightarrow F b$.

- sending each multi-natural transformation $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$ to $(\alpha_\circ)_a := \alpha_a$.

► **Proposition 2.** *The above assignment is a 2-functor $-\circ : \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$.*

It is 2-faithful in the following sense: given two parallel multi-natural transformations

$$\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \alpha \Downarrow \beta \\ \xrightarrow{G} \end{array} \mathcal{V}, \text{ we have } \alpha_\circ = \beta_\circ \implies \alpha = \beta \text{ and so } -\circ \text{ reflects adjunctions.}$$

Proof

To show that the 0-cell mapping is well-defined, we need to show we have a category. Composition of morphisms $a \xrightarrow{f} b \xrightarrow{g} c$ in $\mathcal{U}_\circ$ is their composition in the multi-category, i.e., given by $g \circ f := g \circ (f)$. The multi-category axioms then specialise to:

$$\text{id} \circ (f) = f = f \circ (\text{id}) \quad h \circ (g \circ (f)) = (h \circ (g)) \circ (f)$$

and so we have a category.

To show that the 1-cell mapping is functorial, note that the construction type-checks, hence we have a functor structure, and then immediately validate functoriality:

$$F_\circ(\text{id}) = \text{id} \quad F_\circ(g \circ (f)) = F g \circ (F f)$$

For 2-cells, the resulting transformation $\alpha_{\mathbf{o}} : F_{\mathbf{o}} \rightarrow G_{\mathbf{o}}$ is natural since the multi-naturality condition, when applied to a morphism, recovers the naturality condition for the ordinary transformation.

The 2-faithfulness follows by observing that the data for the multi-natural transformation is preserved by $-\mathbf{o}$. Explicitly, if $\alpha_{\mathbf{o}} = \beta_{\mathbf{o}}$, then for each $a \in \text{Ob}(\mathcal{U})$, we have $\alpha_a = \beta_a$, and that means that $\alpha = \beta$. ■

3 Operads

An *operad* $\mathcal{O}$ consists of:

- a set $\mathcal{O}_n$, whose elements we call *n-ary operations*, for each natural $n \in \mathbb{N}$;
- an identity operation $\text{id} \in \mathcal{O}_1$; and
- a *composed* operation $g \circ (f_1, \dots, f_n) \in \mathcal{O}_{\sum_i n_i}$, for every operation composable pair of operation $g \in \mathcal{O}_n$ and sequence of operations $f_i \in \mathcal{O}_{n_i}$

such that composition is associative and the identity is neutral:

$$\begin{aligned} h \circ (g_1(f_{11}, \dots, f_{1m_1}), \dots, g_n(f_{n1}, \dots, f_{nm_n})) \\ = (h \circ (g_1, \dots, g_n)) \circ (f_{11}, \dots, f_{1m_1}, \dots, f_{n1}, \dots, f_{nm_n}) \quad f \circ (\text{id}, \dots, \text{id}) = f = \text{id} \circ (f) \end{aligned}$$

An operad is then just a 1-object multi-category $\mathcal{O}$, by identifying the n -ary operations $\mathcal{O}_n$ with the n -ary multi-morphisms $\mathcal{O}(\underbrace{\star, \dots, \star}_{n \text{ times}}; \star)$.

The notion of a model may give more intuition about the definition of operads. Let $\mathcal{O}$ be an operad and $\mathcal{U}$ a multi-category. A *model* $\mathcal{A}$ for $\mathcal{O}$ in $\mathcal{U}$ consists of:

- an object $\underline{\mathcal{A}} \in \text{Ob}(\mathcal{U})$
- for each operation $f \in \mathcal{O}_n$, a multi-morphism $\mathcal{A}[[f]] : (\underbrace{\underline{\mathcal{A}}, \dots, \underline{\mathcal{A}}}_{n \text{ times}}) \rightarrow \underline{\mathcal{A}}$

such that composition and identities are preserved:

$$\mathcal{A}[[\text{id}]] = \text{id}_{\underline{\mathcal{A}}} \quad \mathcal{A}[[g \circ (f_1, \dots, f_n)]] = \mathcal{A}[[g]] \circ (\mathcal{A}[[f_1]], \dots, \mathcal{A}[[f_n]])$$

Thus a model of an operad $\mathcal{O}$ in $\mathcal{U}$ is just a multi-functor $\mathcal{A} : \mathcal{O} \rightarrow \mathcal{U}$.

A *homomorphism* $h : \mathcal{A} \rightarrow \mathcal{B}$ between $\mathcal{O}$ -models in $\mathcal{U}$ consists of:

- a morphism $\underline{h} : (\underline{\mathcal{A}}) \rightarrow (\underline{\mathcal{B}})$

such that:

- for all $f \in \mathcal{O}_n$, we have:

$$\underline{h} \circ (\mathcal{A}[[f]]) = \mathcal{B}[[f]](\underline{h}, \dots, \underline{h})$$

Identities and composition of homomorphisms are homomorphisms, and so we have a category $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$ of $\mathcal{O}$ -models in $\mathcal{U}$. We also have that a homomorphism of algebras is

just a multi-natural transformation between them as multi-functors $\mathcal{O} \begin{array}{c} \xrightarrow{\mathcal{A}} \\ \Downarrow \underline{h} \\ \xrightarrow{\mathcal{B}} \end{array} \mathcal{U}$.

We'll conclude with a 2-functor $\mathbf{Mod}(\mathcal{O}, -) : \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$.

- For every multicategory $\mathcal{U}$, let $\mathbf{Mod}(\mathcal{O}, \mathcal{U})$ be the category of $\mathcal{O}$ -models in $\mathcal{U}$.
- For every multi-functor $F: \mathcal{U} \rightarrow \mathcal{V}$, the functor $\mathbf{Mod}(\mathcal{O}, F): \mathbf{Mod}(\mathcal{O}, \mathcal{U}) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathcal{V})$ sends:
 - each $\mathcal{O}$ -model $\mathcal{A}$ in $\mathcal{U}$ to the $\mathcal{O}$ -model:
 - * $F\mathcal{A} := F\mathcal{A}$
 - * $(F\mathcal{A})[[f]]: (F\mathcal{A}, \dots, F\mathcal{A}) \xrightarrow{F(\mathcal{A}[[f]])} F\mathcal{A}$
 - each $\mathcal{O}$ -homomorphism $h: \mathcal{A} \rightarrow \mathcal{B}$ to the $\mathcal{O}$ -homomorphism over $Fh: F\mathcal{A} \rightarrow F\mathcal{B}$.
- For every multi-natural transformation $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$ we have a natural transformation

$$\mathbf{Mod}(\mathcal{O}, \mathcal{U}) \begin{array}{c} \xrightarrow{\mathbf{Mod}(\mathcal{O}, F)} \\ \Downarrow \mathbf{Mod}(\mathcal{O}, \alpha) \\ \xrightarrow{\mathbf{Mod}(\mathcal{O}, G)} \end{array} \mathbf{Mod}(\mathcal{O}, \mathcal{V}) \quad \text{given, for every model } \mathcal{A}, \text{ by:}$$

$$\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}}: (F\mathcal{A}) \xrightarrow{\alpha_{\mathcal{A}}} G\mathcal{A}$$

► **Proposition 3.** *This data defines a 2-functor $\mathbf{Mod}(\mathcal{O}, -): \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$.*

Proof

To see why the action on 1-cells is well-defined and functorial:

- For any $\mathcal{O}$ -model $\mathcal{A}$ in $\mathcal{U}$, view the model as a multi-functor $\mathcal{A}: \mathcal{O} \rightarrow \mathcal{U}$, and note that the model $F\mathcal{A}$ is given by $F\mathcal{A}: \mathcal{O} \xrightarrow{\mathcal{A}} \mathcal{U} \xrightarrow{F} \mathcal{V}$. Therefore it is a model.
- For any $\mathcal{O}$ -model homomorphism $h: \mathcal{A} \rightarrow \mathcal{B}$ in $\mathcal{U}$, view it as a multi-natural transformation $\mathcal{O} \begin{array}{c} \xrightarrow{\mathcal{A}} \\ \Downarrow h \\ \xrightarrow{\mathcal{B}} \end{array} \mathcal{U}$ and note that $\mathbf{Mod}(\mathcal{O}, F)$ sends it to the horizontal composition $\mathcal{O} \begin{array}{c} \xrightarrow{\mathcal{A}} \\ \Downarrow h \\ \xrightarrow{\mathcal{B}} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ = \\ \xrightarrow{F} \end{array} \mathcal{V}$. Therefore:

$$\begin{array}{ccc} \mathcal{O} & \begin{array}{c} \xrightarrow{\mathcal{A}} \\ \Downarrow h_1 \\ \xrightarrow{\mathcal{B}} \end{array} & \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ = \\ \xrightarrow{F} \end{array} \mathcal{V} \\ \downarrow h_2 & & \downarrow \\ \mathcal{O} & \begin{array}{c} \xrightarrow{\mathcal{A}} \\ = \\ \xrightarrow{\mathcal{A}} \end{array} & \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ = \\ \xrightarrow{F} \end{array} \mathcal{V} \end{array}$$

To see why the action on 2-cells is well-defined and functorial:

- Consider any multi-natural transformation $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$. Since the multicategory $\mathcal{O}$ has only one object, $\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}}: F \circ \mathcal{A} \rightarrow G \circ \mathcal{A}$ would completely determine a multi-natural transformation, and it is the multi-natural transformation given by whiskering by $\mathcal{A}$:

$$\mathcal{O} \begin{array}{c} \xrightarrow{\mathcal{A}} \\ = \\ \xrightarrow{\mathcal{A}} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$$

It is therefore natural, i.e., each component is an $\mathcal{O}$ -homomorphism

$$\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}} : \mathbf{Mod}(\mathcal{O}, F)\mathcal{A} = F\mathcal{A} \rightarrow G\mathcal{A} = \mathbf{Mod}(\mathcal{O}, G)\mathcal{A}$$

- To see that $\mathbf{Mod}(\mathcal{O}, \alpha)$ is natural, take any homomorphism $\mathcal{O} \begin{array}{c} \xrightarrow{A} \\ \Downarrow h \\ \xrightarrow{B} \end{array} \mathcal{U}$. The naturality condition then manifests as the two ways to express the horizontal composition at its unique component:

$$\mathcal{O} \begin{array}{c} \xrightarrow{A} \\ \Downarrow h \\ \xrightarrow{A} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$$

- The preservation of vertical and horizontal composition and identities of 2-cells then follows by taking an $\mathcal{O}$ -model, then partitioning these pasting diagrams accordingly:

$$\mathcal{O} \begin{array}{c} \xrightarrow{A} \\ = \\ \xrightarrow{A} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ G \Downarrow \alpha \\ \Downarrow \beta \\ H \end{array} \mathcal{V} \quad \mathcal{O} \begin{array}{c} \xrightarrow{A} \\ = \\ \xrightarrow{A} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F_1} \\ \Downarrow \alpha \\ G_1 \end{array} \mathcal{V} \begin{array}{c} \xrightarrow{F_2} \\ \Downarrow \beta \\ G_2 \end{array} \mathcal{W} \quad \mathcal{O} \begin{array}{c} \xrightarrow{A} \\ = \\ \xrightarrow{A} \end{array} \mathcal{U} \begin{array}{c} \xrightarrow{F} \\ = \\ \xrightarrow{G} \end{array} \mathcal{V}$$

We therefore have a 2-functor. ■

4 Multi-homomorphisms

Let $\mathcal{T}$ be an algebraic theory, and $\langle \mathbb{C}, \Pi \rangle$ a category with chosen finite products. For all objects X, Y in $\mathbb{C}$, and natural number k , define the strength (for the reader monad X^k):

$$\text{str} : X^k \times Y \xrightarrow{(\pi_i \times \text{id})_{i=1}^k} (X \times Y)^k$$

Let $\mathcal{A}_1, \dots, \mathcal{A}_n, \mathcal{B} \in \mathbf{Mod}(\mathcal{T}, \mathbb{C})$ be algebras. A function $f : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ is a *multi- $\mathcal{T}$ -homomorphism* (Hyland and Power, 2006) when, for every term $t \in \mathcal{T}_k$ and $1 \leq i_0 \leq n$, the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{A}_{i_0}^k \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} \mathcal{A}_i & \xrightarrow{\text{str}} & \left(\mathcal{A}_{i_0} \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} \mathcal{A}_i \right)^k & \xrightarrow{\cong} & \left(\prod_{i=1}^n \mathcal{A}_i \right)^k \\ \downarrow \mathcal{A}_{i_0} \llbracket t \rrbracket \times \text{id} & & & & \downarrow f^k \\ & & = & & \mathcal{B}^k \\ & & & & \downarrow \mathcal{B} \llbracket t \rrbracket \\ \mathcal{A}_{i_0} \times \prod_{\substack{1 \leq i \leq n \\ i \neq i_0}} \mathcal{A}_i & \xrightarrow{\cong} & \prod_{i=1}^n \mathcal{A}_i & \xrightarrow{f} & \mathcal{B} \end{array}$$

When $n = 1$, this condition degenerates to the following diagram, which states that $f : \underline{\mathcal{A}} \rightarrow \underline{\mathcal{B}}$ is a $\mathcal{T}$ -homomorphism:

$$\begin{array}{ccc}
\mathcal{A}_{i_0}^k & \xrightarrow{f^k} & \mathcal{B}^k \\
\mathcal{A}_{i_0} \llbracket t \rrbracket \downarrow & = & \downarrow \mathcal{B} \llbracket t \rrbracket \\
\mathcal{A}_{i_0} & \xrightarrow{f} & \mathcal{B}
\end{array}$$

Therefore, multi- $\mathcal{T}$ -homomorphisms generalise $\mathcal{T}$ -homomorphisms:

► **Proposition 4.** *Consider the bijection between 1-ary multi-morphisms $h : (\underline{\mathcal{A}}) \rightarrow \underline{\mathcal{A}}$ and $\mathcal{C}$ -morphisms $h' : \underline{\mathcal{A}} \cong \prod \{\underline{\mathcal{A}}\} \xrightarrow{h} \underline{\mathcal{A}}$. A multi-morphism h is a multi- $\mathcal{T}$ -homomorphism iff the corresponding h' is a $\mathcal{T}$ -homomorphism.*

We will define a multi-category structure $\underline{} : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \rightarrow \langle \mathcal{C}, \Pi \rangle$ over the finite-product multi-category:

- The objects are $\mathcal{T}$ -algebras.
- The multi-morphisms are the multi- $\mathcal{T}$ -homomorphisms.
- The multi-functor $\underline{}$ sends an algebra to its carrier and a multi- $\mathcal{T}$ -homomorphism to its underlying morphism.
- The identities are the identity morphisms.
- Composition is inherited from $\langle \mathcal{C}, \Pi \rangle$.

The next few propositions show that this data defines a multi-category.

► **Proposition 5.** *Let $g : (\mathcal{B}_1, \dots, \mathcal{B}_n) \rightarrow \mathcal{C}$ and $f_i : (\mathcal{A}_{i1}, \dots, \mathcal{A}_{im_i}) \rightarrow \mathcal{B}_i$ be multi- $\mathcal{T}$ -homomorphisms. Then $g \circ (f_1, \dots, f_n) : (\mathcal{A}_{11}, \dots, \mathcal{A}_{nm_n}) \rightarrow \mathcal{C}$ is a multi- $\mathcal{T}$ -homomorphism.*

Proof

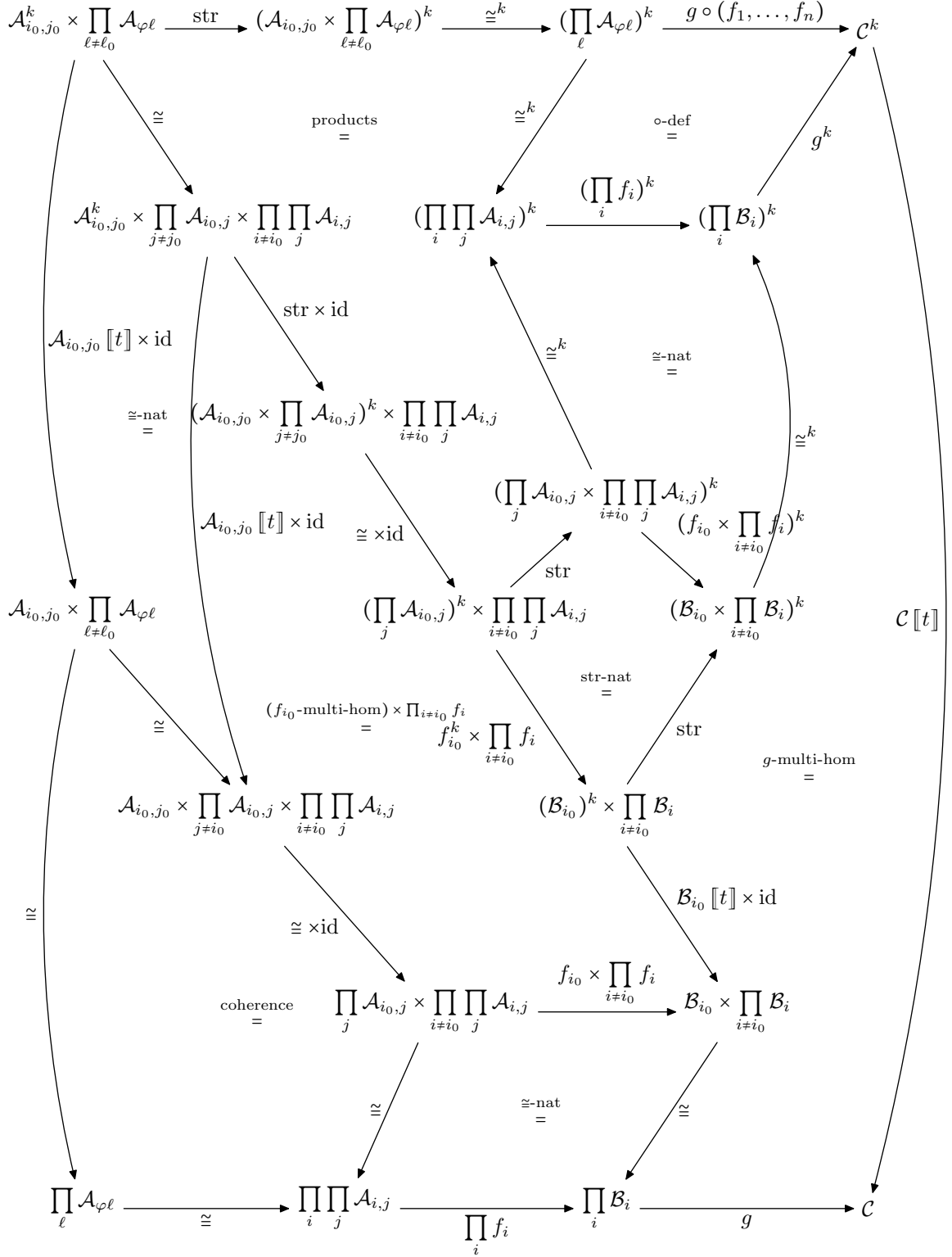
Let $p := \sum_{i=1}^n m_i$ and $\varphi : [p] \xrightarrow{\cong} \coprod_{i=1}^n [m_i]$ be the enumeration from the definition of composition in $\langle \mathcal{C}, \Pi \rangle$ for this case. Take any $1 \leq \ell_0 \leq p$, and let $\langle i_0, j_0 \rangle := \varphi \ell_0$ be its corresponding pair of indices. Take any k -ary term $t \in \mathcal{T}_k$, and calculate as in Fig. 1. ■

► **Proposition 6.** *The data above defines a multi-category $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$ together with a faithful multi-functor $\underline{} : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \rightarrow \langle \mathcal{C}, \Pi \rangle$: $\underline{f_1} = \underline{f_2} \implies f_1 = f_2$ for every parallel pair of multi- $\mathcal{T}$ -homomorphisms $f_1, f_2 : (\mathcal{A}_1, \dots, \mathcal{A}_n) \rightarrow \mathcal{B}$.*

Proof

Identities are multi- $\mathcal{T}$ -homomorphisms by Proposition 4, and composition of multi- $\mathcal{T}$ -homomorphisms is a multi- $\mathcal{T}$ -homomorphism by Proposition 5, and so we have a multi-category structure.

Since composition and identities are inherited from $\langle \mathcal{C}, \Pi \rangle$, we have the structure of multi-functor. This multi-functor structure is faithful since the multi-morphisms in $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$ are those of $\langle \mathcal{C}, \Pi \rangle$ with additional structure. Since faithful multi-functors reflect multi-categories, we conclude. ■



■ **Figure 1** Multi- $\mathcal{T}$ -homomorphism property of the composition

5 Yoneda structure

We'll need universal properties in the 2-category of multi-categories. It might be useful to develop a rich theory of representability inside **MultiCAT**. We'll try to do it using the *Yoneda structure* of Street and Walters (1978).

5.1 Admissible multi-functors

A multi-functor $F : \mathcal{U} \rightarrow \mathcal{V}$ is *admissible* when, for all $a_1, \dots, a_n$ in $\mathcal{U}$ and b in $\mathcal{V}$, the multi-homset $\mathcal{V}(Fa_1, \dots, Fa_n; b) \in \mathbf{Set}$ is small. A multi-category is *admissible* when its identity multi-functor is admissible, i.e., when it is locally small. The admissible multi-functors are closed under-precomposition, i.e., for $\mathcal{U} \xrightarrow{F} \mathcal{V} \xrightarrow{G} \mathcal{W}$, if G is admissible, then $G \circ F$ is also admissible. Indeed, given a_i in $\mathcal{U}$ and c in $\mathcal{W}$, we have:

$$\mathcal{W}(G \circ Fa_1, \dots, G \circ Fa_n; C) = \mathcal{W}(G(Fa_1), \dots, G(Fa_n); C) \in \mathbf{Set}$$

$\begin{array}{c} G \text{ admissibility} \\ \text{for } Fa_1, \dots, Fa_n \\ \downarrow \end{array}$

5.2 Iterated lists

As you can see in from previous sections, manipulating multi-categories in the abstract makes the indexing heavy. Here we develop some notation to manipulate sequences of multi-morphisms ensemble, which is probably related to a construction of a free monoidal category out of a multi-category.

Recall the structure of the list/free monoid monad, given by:

- For each set X , the set $\mathbf{List} X$ of X -sequences $\mathbf{x} := \langle \mathbf{x}_1, \dots, \mathbf{x}_{|\mathbf{x}|} \rangle$, i.e., $|\mathbf{x}|$ is the length of the sequence, and its components in order are $\mathbf{x}_i$ for $1 \leq i \leq n$.

We will also grade the monad, i.e., $\mathbf{List}_n X \subseteq \mathbf{List} X$ are the length n sequences.

- For each $x \in X$, the singleton $\langle x \rangle \in \mathbf{List}_1 X$.
- Given a sequence $\mathbf{x} \in \mathbf{List}_n X$ and, for each $1 \leq i \leq |\mathbf{x}|$, another sequence $\mathbf{y}_i \in \mathbf{List}_{m_i} Y$, we have the concatenated sequence:

$$\biguplus_{i=1}^n \mathbf{y}^i := \langle \mathbf{y}_{11}, \dots, \mathbf{y}_{1m_1}, \dots, \mathbf{y}_{n1}, \dots, \mathbf{y}_{nm_n} \rangle$$

We will also need iterated lists. These too form a monad-like structure, that of *unlabelled, rooted, finitely branching trees of uniform depth*:

- For each set X , the set $\mathbf{Tree} X := \bigcup_{k \in \mathbb{N}} \mathbf{List}^k X$ represents finitely-branching rooted trees whose nodes are unlabelled, and whose leaves, all of which have the same depth, are taken from X .
- For each $x \in X$, we identify the unique rooted tree with 1 leaf at depth 1 labelled by x with the singleton list $\langle x \rangle \in \mathbf{List}^1 X$.

The empty tree, that only has the root node and no leaves, is represented by the empty list.

Note that trees are allowed to have childless *nodes* at non-uniform depths, e.g.: $\langle \langle \rangle, \langle \langle \rangle \rangle, \langle \langle \langle 1 \rangle \rangle \rangle \rangle$ has two nodes at depths 1 and 2 and a leaf at depth 3.

- Given a tree whose leaves are trees, we can substitute the inner trees into the outer tree.

Let $[n] := \{1, \dots, n\}$ be the canonical set of n -many indices. For each depth- k iterated list $\chi \in \text{List}^k X$, let $[\chi] \in \text{List } X$ be its fully flattened sequence.

We denote the set of *branches* of a rooted tree χ by $\text{Branch } \chi$, i.e., sequences of indices in the tree that mark a path from the root to a leaf. The leaf-depths are uniform, we have, for each $\chi \in \text{List}^k X$:

$$\text{Branch } \chi = \coprod_{i_1 \in [\chi]} \coprod_{i_2 \in [\chi_{i_1}]} \cdots \coprod_{i_{k-1} \in [\chi_{i_1, i_2, \dots, i_{k-2}}]} [\chi_{i_1, \dots, i_{k-1}}]$$

Let the *fringe size* of a tree be its number of leaves, i.e., the cardinality of its set of branches:

$$\begin{aligned} \|\chi\| := |\chi| &:= \sum_{i_1 \in [\chi]} \sum_{i_2 \in [\chi_{i_1}]} \cdots \sum_{i_{k-1} \in [\chi_{i_1, i_2, \dots, i_{k-2}}]} |\chi_{i_1, i_2, \dots, i_{k-2}, i_{k-1}}| \\ &= |\text{Branch } \chi| \end{aligned}$$

For example, for $\chi := \langle \langle \langle 1, 2 \rangle, \langle 3, 4 \rangle \rangle, \langle \langle 5 \rangle, \langle 6, 7 \rangle, \langle 8 \rangle \rangle \rangle$ we have $[\chi] = \langle 1, 2, \dots, 8 \rangle$ and:

$$|\chi| = 2 \quad |\chi_1| = 2 \quad |\chi_2| = 3 \quad \|\chi\| := \sum_{i_1 \in [\chi]} \sum_{i_2 \in [\chi_{i_1}]} |\chi_{i_1, i_2}| = (2+2) + (1+2+1) = 8$$

We then define the bijection:

$$\begin{aligned} \langle -, \dots, - \rangle_\chi : \text{Branch } \chi &\xrightarrow{\cong} [\|\chi\|] \\ \langle i_1, \dots, i_k \rangle_\chi &:= \sum_{j_1 < i_1} \|\chi_{j_1}\| + \sum_{j_2 < i_2} \|\chi_{i_1, j_2}\| + \dots + \sum_{j_{k-1} < i_{k-1}} \|\chi_{i_1, i_2, \dots, i_{k-2}, j_{k-1}}\| + i_k \end{aligned}$$

so that $[\chi]_{\langle i_1, \dots, i_k \rangle} = \chi_{i_1, \dots, i_k}$ and $\langle \chi_{\langle 1 \rangle_\chi}^{-1}, \dots, \chi_{\langle \|\chi\|_\chi}^{-1} \rangle$. For example, continuing with the previous χ , we have:

$$\begin{aligned} \langle 1, 2, 1 \rangle &= 0 + 2 + 1 = 3 & [\chi]_3 &= 3 = \chi_{1,2,1} \\ \langle 2, 2, 1 \rangle &= 4 + (1+2) + 1 = 8 & [\chi]_8 &= 8 = \chi_{2,2,1} \end{aligned}$$

Given a list $\mathbf{a} \in \text{List } A$ over some set A , an *ordered partition* of $\mathbf{a}$ is a list-of-lists $\pi \in \text{List}^2 A$ such that $\mathbb{H}_{i=1}^{|\pi|} \pi_i = \mathbf{a}$. Similarly, an *iterated ordered partition* of $\mathbf{a}$ is a tree $\pi \in \text{Tree } A$ such that $[\pi] = \mathbf{a}$.

5.3 The multi-category of presheaves

Let $\mathcal{U}$ be a multi-category. We will define a *category* $\mathcal{U}^{\text{op}}$ so that our presheaves will be functors $F : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$:

- $\text{Ob } (\mathcal{U}^{\text{op}}) := \text{List Ob } (\mathcal{U})$ are finite-sequences $\mathbf{a} = \langle \mathbf{a}_1, \dots, \mathbf{a}_n \rangle$ of $\mathcal{U}$ -objects.
- Morphisms $f \in \mathcal{U}^{\text{op}}(\mathbf{a}, \mathbf{b})$, which we will write as “ $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$ ”, consist of:
 - an ordered partition $\pi_f := \langle \pi_1, \dots, \pi_n \rangle \in \text{List}^2 \text{Ob } (\mathcal{U})$, $\vec{b}$, i.e.: $\mathbf{b} = \mathbb{H}_{i=1}^n \pi_i$.

- for each $1 \leq i \leq n$, a multi-morphism $f_i : \mathbf{a}_i \leftarrow (\mathbf{b}_i)$. We can depict a morphism like so:

$$\begin{array}{c}
 \begin{array}{c} \mathbf{a}_1 \longleftarrow f_1 \end{array} \begin{array}{c} \nearrow \pi_{11} \\ \searrow \pi_{1|\mathbf{b}_1|} \end{array} \begin{array}{c} = \\ \mathbf{b}_{\langle 1,1 \rangle} \end{array} \\
 \vdots \\
 \begin{array}{c} \mathbf{a}_n \longleftarrow f_n \end{array} \begin{array}{c} \nearrow \pi_{n1} \\ \searrow \pi_{n|\mathbf{b}_n|} \end{array} \begin{array}{c} = \\ \mathbf{b}_{\langle n,1 \rangle} \end{array}
 \end{array}$$

- The identity $\text{id}_{\bar{a}}^{[\mathcal{U}^{\text{op}}]}$ consists of:
 - * $\pi_{\text{id}_{\bar{a}}} := \langle \langle a_1 \rangle, \dots, \langle a_n \rangle \rangle$, the partition into singletons; and
 - * $(\text{id}_{\bar{a}})_i := \text{id}_{a_i} : a_i \leftarrow (a_i)$.
 - To compose morphisms $\mathbf{a} \xleftarrow{f} \mathbf{b} \xleftarrow{g} \mathbf{c}$, use the multiplication of the **Tree** monad to get an iterated partition of depth 3, and then flatten all the depth 1 iterated lists into lists. Then, compose the multi-morphisms to get a morphism $f \circ g : \mathbf{a} \leftarrow \mathbf{c}$ (note the order of arguments in this notation). Letting $|\mathbf{a}| =: n$ and $|\pi_{f,i}| =: m_i$:
 - * $\pi_{f \circ g} := \langle \langle \pi_{g_{\langle 1,1 \rangle}} \# \dots \# \pi_{g_{\langle 1,m_1 \rangle}} \rangle, \dots, \langle \pi_{g_{\langle n,1 \rangle}} \# \dots \# \pi_{g_{\langle n,m_n \rangle}} \rangle \rangle$
 - * $(f \circ g)_i := f_i \circ (g_{\langle i,1 \rangle}, \dots, g_{\langle i,m_i \rangle})$.
- We can visualise this operation so, letting $k_{i,j} =: |\pi_{g_i, j}|$:

$$\begin{array}{c}
 \begin{array}{c} \mathbf{a}_1 \longleftarrow f_1 \end{array} \begin{array}{c} \nearrow \pi_{11} \\ \searrow \pi_{1|\mathbf{b}_1|} \end{array} \begin{array}{c} = \\ \mathbf{b}_{\langle 1,1 \rangle} \end{array} \begin{array}{c} \nearrow g_{\langle 1,1 \rangle} \\ \searrow g_{\langle 1,m_1 \rangle} \end{array} \begin{array}{c} = \\ \mathbf{c}_{\langle 1,1,k_{11} \rangle} \end{array} \\
 \vdots \\
 \begin{array}{c} \mathbf{a}_n \longleftarrow f_n \end{array} \begin{array}{c} \nearrow \pi_{n1} \\ \searrow \pi_{n|\mathbf{b}_n|} \end{array} \begin{array}{c} = \\ \mathbf{b}_{\langle n,1 \rangle} \end{array} \begin{array}{c} \nearrow g_{\langle n,1 \rangle} \\ \searrow g_{\langle n,m_n \rangle} \end{array} \begin{array}{c} = \\ \mathbf{c}_{\langle n,1,k_{n1} \rangle} \end{array}
 \end{array}$$

► **Lemma 7.** *This data defines a category $\mathcal{U}^{\text{op}}$.*

Proof

We inherit the properties from the monad laws and the multi-category axioms. ■

A *presheaf* over $\mathcal{U}$ is a functor $F : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$.

► **Example 6.** Let $\mathcal{U}$ be a locally small multi-category. For each $a \in \text{Ob}(\mathcal{U})$, we have a presheaf $\mathbf{y}_{\mathcal{U}}[a] : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$ called the *presheaf represented by a* :

$$\mathbf{y}_{\mathcal{U}}[a] \mathbf{b} := \mathcal{U}(\mathbf{b}; a) \in \mathbf{Set} \quad \mathbf{y}_{\mathcal{U}}[a] (\mathbf{b} \xleftarrow{f} \mathbf{c}) : g \in \mathcal{U}(\mathbf{b}; a) \mapsto g \circ (f_1, \dots, f_{|\mathbf{b}|}) \in \mathcal{U}(\mathbf{c}; a)$$

The functoriality:

$$\mathbf{y}_{\mathcal{U}}[a] (\mathbf{b} \xleftarrow{f} \mathbf{c} \xleftarrow{g} \mathbf{d}) = \mathbf{y}_{\mathcal{U}}[a] (g) \circ \mathbf{y}_{\mathcal{U}}[a] (f) \quad \mathbf{y}_{\mathcal{U}}[a] (\text{id}_{\mathbf{b}}) = \text{id}_{\mathbf{y}_{\mathcal{U}}[a] \mathbf{b}}$$

follows from the multi-category axioms and the definition of $\mathcal{U}^{\text{op}}$. ◀

Let $F_1, \dots, F_n, G : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$ be presheaves. A *natural multi-transformation* $\alpha : (F_1, \dots, F_n) \rightarrow G$ consists of:

- For every sequence $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$ and length- n ordered partition π of $\mathbf{a}$, a function:

$$\alpha_{\pi} : \prod_{i=1}^n F_i \pi_i \rightarrow G \mathbf{a}$$

such that:

- For every $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$ and for every length- n ordered partition $\varphi = \langle \varphi_i : \pi_i \leftarrow \pi'_i \rangle_{i=1}^n$ of the sequence f :

$$\begin{array}{ccc} \prod_i F_i \pi_i & \xrightarrow{\alpha_{\pi}} & G \mathbf{a} \\ \prod_i F \varphi_i \downarrow & = & \downarrow G f \\ \prod_i F_i \pi'_i & \xrightarrow{\alpha_{\pi'}} & G \mathbf{b} \end{array}$$

► **Example 7.** Let $g : (\mathbf{a}) \rightarrow b$ be a multi-morphism in $\mathcal{U}$. We define a natural multi-transformation, where $n := |\mathbf{a}|$:

$$\begin{aligned} \mathbf{y}_{\mathcal{U}} g : (\mathbf{y}_{\mathcal{U}} \mathbf{a}_1, \dots, \mathbf{y}_{\mathcal{U}} \mathbf{a}_n) &\rightarrow \mathbf{y}_{\mathcal{U}} b \\ (\mathbf{y}_{\mathcal{U}} g)_{\pi} : \langle \pi_1 \xrightarrow{u_1} \mathbf{a}_1, \dots, \pi_n \xrightarrow{u_n} \mathbf{a}_n \rangle &\mapsto g \circ (u_1, \dots, u_n) \end{aligned}$$

To see that it is natural, take any $f : \mathbf{c} \leftarrow \mathbf{d}$ in $\mathcal{U}$, and an n -ary ordered partition of it φ , and chase elements around the naturality diagram:

$$\begin{array}{ccc} \langle \pi_i \xrightarrow{u_i} \mathbf{a}_i \rangle_{i=1}^n & \xrightarrow{(\mathbf{y}g)_{\pi}} & g \circ (u_1, \dots, u_n) \\ \prod_i \mathbf{y} \mathbf{a}_i \varphi_i \downarrow & & \downarrow \mathbf{y} b g \\ \langle \pi'_i \xrightarrow{\varphi_i} \pi_i \xrightarrow{u_i} \mathbf{a}_i \rangle_{i=1}^n & \xrightarrow{(\mathbf{y}g)_{\pi'}} & f \circ (u_1 \circ (\varphi_{11}, \dots, \varphi_{1m_1}), \dots, u_n \circ (\varphi_{n1}, \dots, \varphi_{nm_n})) \end{array}$$

$\begin{array}{c} (f \circ (u_1, \dots, u_n)) \circ (h_1, \dots, h_p) \\ \parallel \leftarrow \begin{array}{l} \text{multi-cat. associativity} \\ \text{since } \langle h_{\ell} \rangle_{\ell=1}^p = \langle \varphi_{ij} \rangle_{i=1, j=1}^{n, m_n} \end{array} \end{array}$

and so $\mathbf{y}g$ is a natural multi-transformation. $\blacktriangleleft$

The multi-category $\widehat{\mathcal{U}}$ of *presheaves over $\mathcal{U}$* consists of this data:

- Objects are presheaves $F : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$.
- Multi-morphisms $\alpha : (F_1, \dots, F_n) \rightarrow G$ are the natural multi-transformations.
- The identity multi-transformation $\text{id}_F : (F) \rightarrow F$ has as component at each unary partition $\pi = \langle \mathbf{a} \rangle$ of $\mathbf{a}$ the canonical isomorphism:

$$(\text{id}_F)_{\langle \mathbf{a} \rangle} : \prod \{F\mathbf{a}\} \xrightarrow{\cong} F\mathbf{a}$$

- To define composition, take composable natural multi-transformations

$$\alpha = \langle (F_i) \rightarrow G_i \rangle_{i=1}^n \quad \text{and} \quad \beta : (G) \rightarrow H \quad \text{where:} \quad \mathbf{F} \in \text{List}^2(\mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}), \mathbf{G} \in \text{List}^1(\mathcal{U}^{\text{op}} \rightarrow \mathbf{Set})$$

letting as usual $p := \|\mathbf{F}\|$, $n := |\mathbf{G}|$, and $m_i := |\mathbf{F}_i|$. Given a p -ary ordered partition π of some $\mathbf{a}$, define these two auxiliary partitions:

- $\pi_{\langle i, * \rangle} := \langle \pi_{\langle i, j \rangle} \rangle_{j=1}^{m_i}$
- $\pi_{\langle i, \# \rangle} := \#_{j=1}^{m_i} \pi_{\langle i, j \rangle} = \lfloor \pi'_i \rfloor$
- So $\alpha_{i, \pi_{\langle i, * \rangle}} : \prod_{j=1}^{m_i} \mathbf{F}_{\langle i, j \rangle} \rightarrow \mathbf{G}_i \pi_{\langle i, \# \rangle}$.

It might easier to visualise how they relate to each other by aligning their fringes:

$$\begin{aligned} \pi &= \langle \pi_{\langle 1, 1 \rangle}, \dots, \pi_{\langle 1, m_1 \rangle}, \dots, \pi_{\langle n, 1 \rangle}, \dots, \pi_{\langle n, m_n \rangle} \rangle \\ \pi_{\langle *, * \rangle} &:= \langle \langle \pi_{\langle 1, 1 \rangle}, \dots, \pi_{\langle 1, m_1 \rangle} \rangle, \dots, \langle \pi_{\langle n, 1 \rangle}, \dots, \pi_{\langle n, m_n \rangle} \rangle \rangle \\ \pi_{\langle *, \# \rangle} &:= \langle \pi_{\langle 1, 1 \rangle} \# \dots \# \pi_{\langle 1, m_1 \rangle}, \dots, \pi_{\langle n, 1 \rangle} \# \dots \# \pi_{\langle n, m_n \rangle} \rangle \end{aligned}$$

We define the π -th component of the composition:

$$(\beta \circ (\alpha_1, \dots, \alpha_n))_{\pi} : \prod_{\ell=1}^p \mathbf{F}_{\ell} \pi_{\ell} \xrightarrow{\cong} \prod_{i=1}^n \prod_{j=1}^{m_i} \mathbf{F}_{\langle i, j \rangle} \pi_{\langle i, j \rangle} \xrightarrow{\prod_{i=1}^n \alpha_{i, \pi_{\langle i, * \rangle}}} \prod_{i=1}^n \mathbf{G}_i \pi_{\langle i, \# \rangle} \xrightarrow{\beta_{\pi_{\langle *, \# \rangle}}} H\mathbf{a}$$

► **Proposition 8.** *These data define a multi-category $\widehat{\mathcal{U}}$.*

Proof

First, we need to show the identities and composition are well-defined, i.e., natural. For the identity, take any unary partition $\varphi = \langle f \rangle$ of a morphism $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$, and calculate:

$$\begin{array}{ccc} \prod \{F\mathbf{a}\} & \xrightarrow[\cong]{\text{id}_{F, \langle \mathbf{a} \rangle}} & F\mathbf{a} \\ \downarrow & \cong\text{-nat} & \downarrow Ff \\ \prod \{F\mathbf{b}\} & \xrightarrow[\cong]{\text{id}_{F, \langle \mathbf{b} \rangle}} & F\mathbf{b} \end{array}$$

Turning to the well-definedness of composition, continue with the same notation and take a p -ary partition $\varphi = \langle \varphi_{\ell} : \pi_{\ell} \leftarrow \sigma_{\ell} \rangle$ of some morphism $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$. The crucial step is to repeat the same partitioning scheme for φ as well:

- $\varphi_{\langle i, * \rangle} := \langle \varphi_{\langle i, j \rangle} : \pi_{\langle i, j \rangle} \leftarrow \sigma_{\langle i, j \rangle} \rangle_{j=1}^{m_i}$;
- $\varphi_{\langle i, \# \rangle} := \#_{j=1}^{m_i} \varphi_{\langle i, j \rangle} : \pi_{\langle i, \# \rangle} \leftarrow \sigma_{\langle i, \# \rangle}$ so that $\varphi_{\langle i, * \rangle}$ is a partition of $\varphi_{\langle i, \# \rangle}$;

and we can now calculate the naturality condition of the composition:

$$\begin{array}{ccc}
\prod_{\ell=1}^p \mathbf{F}_\ell \pi_\ell & \xrightarrow{(\beta \circ (\alpha_1, \dots, \alpha_n))_\pi} & Ha \\
\downarrow \cong & \searrow \text{o-def} = & \downarrow \beta_{\pi(*, \#)} \\
\prod_i \prod_j \mathbf{F}_{\langle i, j \rangle} \pi_{\langle i, j \rangle} & \xrightarrow{\prod_i \alpha_{i, \pi_{\langle i, * \rangle}}} & \prod_{i=1}^n \mathbf{G}_i \pi_{\langle i, \# \rangle} \\
\downarrow \cong \text{-nat} = & \downarrow \Pi_i (\alpha_i \text{-nat.}) = & \downarrow \prod_i \mathbf{G}_i \varphi_{\langle i, \# \rangle} \beta \text{-nat} = \\
\prod_i \prod_j \mathbf{F}_{\langle i, j \rangle} \sigma_{\langle i, j \rangle} & \xrightarrow{\prod_i \alpha_{i, \sigma_{\langle i, * \rangle}}} & \prod_{i=1}^n \mathbf{G}_i \sigma_{\langle i, \# \rangle} \\
\downarrow \cong & \searrow \text{o-def} = & \downarrow \beta_{\sigma(*, \#)} \\
\prod_{\ell=1}^p \mathbf{F}_\ell \sigma_\ell & \xrightarrow{(\beta \circ (\alpha_1, \dots, \alpha_n))_\sigma} & Hb
\end{array}$$

$\prod_{\ell=1}^p \mathbf{F}_\ell \varphi_\ell$ (left side), Hf (right side)

The neutrality of the identities w.r.t. composition follow immediately:

$$\begin{array}{ccc}
(\alpha \circ (\text{id}_{\mathbf{F}_1}, \dots, \text{id}_{\mathbf{F}_n}))_\pi & & (\text{id}_G \circ (\alpha))_\pi \\
\downarrow \text{o-def} = & & \downarrow \text{o-def} = \\
\prod_{\ell=1}^n \prod_{\ell=1}^n \{\mathbf{F}_\ell \pi_\ell\} & \xrightarrow{\prod_{\ell=1}^n \alpha_\pi} & \prod_{\ell=1}^n \prod_{\ell=1}^n \{\mathbf{F}_\ell \pi_\ell\} \\
\downarrow \text{products} = & & \downarrow \text{products} = \\
\prod_{\ell=1}^n \mathbf{F}_\ell \pi_\ell & \xrightarrow{\alpha_\pi} & \prod_{\ell=1}^n \mathbf{F}_\ell \pi_\ell \\
\downarrow \text{id} = & & \downarrow \text{id} = \\
Ga & & Ga
\end{array}$$

To prove that composition is associative, we'll first dwell on the partitioning scheme we used to define composition.

We can organise the data in a sequence of d -many composable multi-transformations:

$$\mathbf{F}^{d+1} \xrightarrow{\alpha^d} \dots \xrightarrow{\alpha^1} \mathbf{F}^1 \xrightarrow{\alpha^0} F^0 \quad ((*))$$

as follows. We consider a rooted tree t of uniform leaf-depth $d+1$. We label each leaf, say in branch $\langle i_1, \dots, i_{d+1} \rangle$ in the tree, with a single presheaf $\mathbf{F}_{\langle i_1, \dots, i_{d+1} \rangle}^{d+1}$. Each node of depth $1 \leq s \leq d$ in the tree, say on the path $\langle i_1, \dots, i_s \rangle$ is also labelled with a presheaf $\mathbf{F}_{i_1, \dots, i_s}^s$. Given these labels, we also annotate each node, that has, say, $k_{\langle i_1, \dots, i_s \rangle}$ children, with a natural multi-transformation $\alpha_{\langle i_1, \dots, i_s \rangle}^s : \left(\left(\mathbf{F}_{i_1, \dots, i_{s+1}}^{s+1} \right)_{i=1}^{k_{\langle i_1, \dots, i_s \rangle}} \right) \rightarrow \mathbf{F}_{i_1, \dots, i_s}^s$.

The bijection between the set of length- s branches and the set of nodes and leaves at level s allows us to define the lists $\mathbf{F}^s$ and α^s to make the notation in $(*)$ precise. After composing these multi-transformations in some order, we get a multi-transformation:

$$\bigcirc \alpha : (\mathbf{F}^{s+1}) \rightarrow \mathbf{F}^0$$

Letting $p := |\mathbf{F}^{s+1}|$, when we take any $\mathbf{a}$ and a p -ary partition of $\mathbf{a}$, we can extend the tree t by another level (turning all the previous leaves into nodes but without the multi-transformation label), and the leaves in the newly added level are labelled by the objects in $\mathbf{a}$. It is this tree, its labels, and its branches and paths that we are manipulating with our notation.

Given a tree (with any sort of labelling) of uniform leaf depth d , we denote by $t_{\langle i_1, \dots, i_s, *, \dots, * \rangle}$ the sub-tree rooted at node $\langle i_1, \dots, i_s \rangle$. It's perhaps convenient to pad the remaining components up to a length- d sequence to spot typos. We denote by:

$$t_{\langle i_1, \dots, i_s, *, \dots, *, \#^*, \dots, *, \#^*, \dots, *, \#^*, \dots, *, \#^*, \dots, * \rangle}$$

$\uparrow \quad \uparrow \quad \uparrow$
 pos. j_1 pos. $j_2 \dots$ pos. j_r

the sub-tree rooted at $\langle i_1, \dots, i_s \rangle$, whose levels $j_1, \dots, j_r$ were flattened. This notation is consistent with the notation we used for composition and in the previous parts of the proof, apart from identifying the partition π with its depth-3 iterated list. The crux of our proof is a kind of substitution lemma/monad-like associativity law that means the various iterations on this notation are consistent with each other, for example:

$$(t_{i_1, i_2, *, \#^*, \#^*, \#^*})_{j_1, \langle j_2, j_3 \rangle, \#} = t_{i_1, i_2, j_1, j_2, j_3, \#^*, \#}$$

Back to proving associativity. Consider 3 levels of composable multi-transformations:

$$\mathbf{F} \xrightarrow{\alpha} \mathbf{G} \xrightarrow{\beta} \mathbf{H} \xrightarrow{\gamma} \mathbf{K}$$

i.e.:

- $\gamma : \left(\langle \mathbf{H}_{\langle i \rangle} \rangle_{i=1}^{k_{\langle \rangle}} \right) \rightarrow \mathbf{K}$
- $\beta_{\langle i \rangle} : \left(\langle \mathbf{G}_{\langle i, j \rangle} \rangle_{j=1}^{k_{\langle i \rangle}} \right) \rightarrow \mathbf{H}_{\langle i \rangle}$, for all $1 \leq i \leq k_{\langle \rangle}$.
- $\alpha_{\langle i, j \rangle} : \left(\langle \mathbf{F}_{\langle i, j, r \rangle} \rangle_{r=1}^{k_{\langle i, j \rangle}} \right) \rightarrow \mathbf{G}_{\langle i, j \rangle}$, for all $1 \leq i \leq k_{\langle \rangle}$ and $1 \leq j \leq k_{\langle i \rangle}$.

Let:

- $f := |\mathbf{F}|$
- $f_{\langle i, j \rangle} := |\mathbf{F}_{\langle i, j, * \rangle}|$
- $g := |\mathbf{G}|$
- $g_i := \|\mathbf{G}_{\langle i, * \rangle}\| = k_{\langle i \rangle}$
- $h := |\mathbf{H}| = k_{\langle \rangle}$

Consider any f -ary partition π of some $\mathbf{a}$, and calculate:

For identity multimorphisms $\text{id}_a : (a) \rightarrow a$, there is only one 1-partition of any sequence $\mathbf{b}$, and we have:

$$(\mathbf{y}\text{id})_{(\mathbf{b})_{i=1}^1} \langle u : (\mathbf{b}) \rightarrow a \rangle_{i=1}^1 \mapsto \text{id} \circ (u) = u = (\text{id}_{\mathbf{y}_a})_{(\mathbf{b})} \langle u \rangle$$

and so $\mathbf{y}\text{id} = \text{id}_{\mathbf{y}_a}$. Taking composable multi-morphisms $g : (\mathbf{c}) \rightarrow d$ and $f_i : (\mathbf{b}_i) \rightarrow \mathbf{c}_i$, $i = 1, \dots, n := |\mathbf{c}|$, take any p -ary partition π of some $\mathbf{a}$, where $p := \|\mathbf{b}\|$, and chase an arbitrary element of $\prod_{\ell=1}^p \mathbf{y}\mathbf{b}_\ell \pi_\ell$, where $m_i := |\mathbf{b}_i|$:

$$\begin{array}{ccc} \left\langle \pi_\ell \xrightarrow{u_\ell} \mathbf{b}_\ell \right\rangle_{\ell=1}^p & \xrightarrow{\cong} & \left\langle \left\langle u_{\langle i,j \rangle} \right\rangle_{j=1}^{m_i} \right\rangle_{i=1}^n \\ \downarrow (\mathbf{y}(g \circ \langle f_i \rangle_{i=1}^n))_\pi & & \downarrow \prod_{i=1}^n (\mathbf{y}f_i)_{\pi_{\langle i,* \rangle}} \\ (g \circ \langle f_i \rangle_{i=1}^n) \circ \langle u_\ell \rangle_{\ell=1}^p & \xrightarrow{\text{composition associativity}} & g \circ \left\langle f_i \circ \left\langle u_{\langle i,j \rangle} \right\rangle_{j=1}^{m_i} \right\rangle_{i=1}^n \\ & \xleftarrow{(\mathbf{y}g)_{\pi_{\langle *,* \rangle}}} & \left\langle f_i \circ \left\langle u_{\langle i,j \rangle} \right\rangle_{j=1}^{m_i} \right\rangle_{i=1}^n \end{array}$$

and so $\mathbf{y}(g \circ \langle f_i \rangle_i) = \mathbf{y}g \circ \langle \mathbf{y}f_i \rangle_i$. ■

► **Lemma 11.** Yoneda for multi-categories. Let $\mathcal{U}$ be a locally-small multi-category, $\mathbf{a}$ an n -ary sequence of objects in $\mathcal{U}$, and $F \in \widehat{\mathcal{U}}$ a presheaf. We have a bijection:

$$\Upsilon_{\mathcal{U}, F, \mathbf{a}} : \widehat{\mathcal{U}}(\mathbf{y}\mathbf{a}_1, \dots, \mathbf{y}\mathbf{a}_n; F) \xrightarrow{\cong} F\mathbf{a}$$

given by:

$$\begin{aligned} \Upsilon_{\mathcal{U}, F, \mathbf{a}}(\langle \mathbf{y}\mathbf{a}_i \rangle_{i=1}^n \xrightarrow{\alpha} F) &:= \alpha_{\langle \mathbf{a}_i \rangle_{i=1}^1} \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n \in F\mathbf{a} \\ (\Upsilon_{\mathcal{U}, F, \mathbf{a}}^{-1}x)_\pi &: \left\langle \pi_i \xrightarrow{u_i} \mathbf{a}_i \right\rangle_{i=1}^n \mapsto F(u_1, \dots, u_n)x \end{aligned}$$

Proof

As usual, the proof is to just do it.

To show well-definedness, i.e., that $\Upsilon^{-1}x : (\mathbf{y}\mathbf{a}_1, \dots, \mathbf{y}\mathbf{a}_n) \rightarrow F$ is natural, take any morphism $\mathbf{b} \xleftarrow{f} \mathbf{c}$ in $\mathcal{U}$, an n -ary partition $\langle \pi_i \xleftarrow{\varphi_i} \pi'_i \rangle_{i=1}^n$ of it, let $m_i := |\pi_i|$, and chase an arbitrary element of $\prod_{i=1}^n \mathbf{y}\mathbf{a}_i \pi_i$:

$$\begin{array}{ccc} \left\langle (\pi_i) \xrightarrow{u_i} \mathbf{a}_i \right\rangle_{i=1}^n & \xrightarrow{(\Upsilon^{-1}x)_\pi} & F(u_1, \dots, u_n)x \\ \downarrow \prod_{i=1}^n \mathbf{y}\mathbf{a}_1 \varphi_i & & \downarrow Ff \\ \left\langle u_i \circ \langle \varphi_{ij} \rangle_{j=1}^{m_i} : (\pi'_i \rightarrow \mathbf{a}_i) \right\rangle_{i=1}^n x & \xrightarrow{(\Upsilon^{-1}x)_{\pi'}} & F \left\langle u_i \circ \langle \varphi_{\langle i,j \rangle} \rangle_{j=1}^{m_i} \right\rangle_{i=1}^n x \\ & \xleftarrow{\text{functoriality}} & F \left\langle \langle \varphi_{\langle i,j \rangle} \rangle_{j=1}^{m_i} \right\rangle_{i=1}^n \circ F \langle u_i \rangle_{i=1}^n x \end{array}$$

|| $\leftarrow \varphi$ partitions f

To show $\Upsilon \circ \Upsilon^{-1} = \text{id}$, take any $x \in F\mathbf{a}$:

$$\begin{array}{c} \text{functoriality} \\ \downarrow \\ \Upsilon(\Upsilon^{-1}x) = F(\text{id}, \dots, \text{id})(x) = x \end{array}$$

For the converse, take any natural multi-transformation $\alpha : (\mathbf{y}\mathbf{a}_1, \dots, \mathbf{a}_n) \rightarrow F$. We need to show that, for every n -ary partition π of some $\mathbf{b}$ and $\pi_i \xrightarrow{u_i} \mathbf{a}_i$:

$$\alpha_\pi(u_1, \dots, u_n) = (\Upsilon^{-1}(\Upsilon\alpha))_\pi(u_1, \dots, u_n) = F(u_1, \dots, u_n) \left(\alpha_{\langle \mathbf{a}_i \rangle_{i=1}^n} (\text{id}_{\mathbf{a}_1}, \dots, \text{id}_{\mathbf{a}_n}) \right)$$

We do so by chasing the element $\langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n \in \prod_{i=1}^n \mathbf{y}\mathbf{a}_i \langle \mathbf{a}_i \rangle_{j=1}^1$ around α 's naturality diagram for the partition $\langle \langle \mathbf{a}_i \rangle_{j=1}^1 \xleftarrow{u_i} \pi_i \rangle_{i=1}^n$ of the morphism $u : \mathbf{a} \xleftarrow{(u_1, \dots, u_n)} \mathbf{b}$ in $\mathcal{U}$:

$$\begin{array}{ccc} \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n & \xrightarrow{\alpha_{\langle \mathbf{a}_i \rangle_{j=1}^1}^n} & \alpha_{\langle \mathbf{a}_i \rangle_{j=1}^1}^n \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n \\ \downarrow \prod_{i=1}^n \mathbf{y}\mathbf{a}_i u_i & & \downarrow F u \\ \langle u_i \rangle_{i=1}^n & \xrightarrow{\alpha_\pi} & \alpha_\pi(\langle u_i \rangle_{i=1}^n) \end{array} \quad \begin{array}{c} F \langle u_i \rangle_{i=1}^n (\alpha_{\langle \mathbf{a}_i \rangle_{j=1}^1}^n \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n) \\ \parallel \leftarrow \text{qed} \end{array}$$

and so $\Upsilon^{-1} \circ \Upsilon = \text{id}$. ■

We call both this result and the bijection Υ the Yoneda lemma.

► **Lemma 12.** Yoneda embedding. *The Yoneda multi-functor $\mathbf{y}_{\mathcal{U}} : \mathcal{U} \rightarrow \widehat{\mathcal{U}}$ is fully-faithful.*

Proof

Take any $\mathbf{a}$ and b , and consider the Yoneda lemma for the representable presheaf $F := \mathbf{y}b$. Take any $f \in \mathcal{U}(\mathbf{a}; b) = \mathbf{y}b\mathbf{a}$. For every partition π of every n -tuple $\mathbf{c}$ and $\langle u_i : \pi_i \rightarrow \mathbf{a}_i \rangle_{i=1}^n$, we have:

$$(\Upsilon_{\mathcal{U}, \mathbf{y}b, \mathbf{a}}^{-1} f)_\pi \langle u_i \rangle_{i=1}^n = \mathbf{y}b(u_1, \dots, u_n) f = f \circ (u_1, \dots, u_n) = (\mathbf{y}f)_\pi \langle u_i \rangle_{i=1}^n$$

and so $\Upsilon_{\mathcal{U}, \mathbf{y}b, \mathbf{a}}^{-1} f = \mathbf{y}_{\mathcal{U}} f$, i.e., $\Upsilon_{\mathcal{U}, \mathbf{y}b, \mathbf{a}}^{-1}$ is the multi-functorial action of $\mathbf{y}$ on the multi-homset $\mathcal{U}(\mathbf{a}; b)$. Since the Yoneda lemma is bijective, this functorial action is bijective, i.e., $\mathbf{y}_{\mathcal{U}}$ is fully-faithful. ■

► **Lemma 13.** Naturality of the Yoneda Lemma. *The Yoneda lemma $\Upsilon : \widehat{\mathcal{U}}(\mathbf{y}[\mathbf{a}], F) \xrightarrow{\cong} F\mathbf{a}$ is natural in the following senses:*

■ *Naturally in $\mathbf{a} \in \mathcal{U}^{\text{op}}$, i.e. for all $f : \mathbf{a} \leftarrow \mathbf{b}$:*

$$Ff(\Upsilon(\mathbf{y}[\mathbf{a}] \xrightarrow{\alpha} F)) = \Upsilon(\mathbf{y}[\mathbf{b}] \xrightarrow{\mathbf{y}[f]} \mathbf{y}[\mathbf{a}] \xrightarrow{\alpha} F)$$

■ *Multi-naturally in $F \in \widehat{\mathcal{U}}$, i.e. for all $(F_1, \dots, F_n) \xrightarrow{\beta} G$ and partition π of $\mathbf{a}$:*

$$\beta_\pi \left(\Upsilon(\mathbf{y}[\pi_i] \xrightarrow{\alpha_i} F_i) \right)_{i=1}^n = \Upsilon \left(\mathbf{y}[\mathbf{a}] \xrightarrow{\beta \circ \langle \alpha_i \rangle_{i=1}^n} G \right)$$

We can recover these from the general theory of Yoneda structures, but we use them to establish the Yoneda structure axioms.

Proof

In both cases, employ mindless calculation. For naturality:

$$\begin{aligned}
Ff(\Upsilon(\mathbf{y}[\mathbf{a}] \xrightarrow{\alpha} F)) &= Ff\left(\alpha_{\langle \mathbf{a}_i \rangle_{j=1}^1}^n \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n\right) \xrightarrow{\alpha\text{-naturality}} \alpha_{\langle \mathbf{b}_{\langle i, * \rangle} \rangle_{i=1}^n} \left\langle (\mathbf{y} \mathbf{a}_i f_i) \langle \text{id}_{\mathbf{a}_i} \rangle_{j=1}^1 \right\rangle_{i=1}^n \\
&\xrightarrow{\text{identities}} \alpha_{\langle \mathbf{b}_{\langle i, * \rangle} \rangle_{i=1}^n} \left\langle (\mathbf{y} f_i)_{\langle \langle \mathbf{b}_{\langle i, j \rangle} \rangle_{k=1}^1}^1 \right\rangle_{j=1}^{|\mathbf{b}_{\langle i, * \rangle}|} \langle \text{id}_{\mathbf{b}_{\langle i, j \rangle}} \rangle_{j=1}^{|\mathbf{b}_{\langle i, * \rangle}|} \Bigg\rangle_{i=1}^n \\
&\xrightarrow{\text{o-def}} (\alpha \circ \langle \mathbf{y} f_i \rangle_{i=1}^n)_{\langle \langle \mathbf{b}_\ell \rangle_{j=1}^1 \rangle_{\ell=1}^{\|\mathbf{b}\|}} \langle \text{id}_{\mathbf{b}_\ell} \rangle_{\ell=1}^{\|\mathbf{b}\|} = \Upsilon(\alpha \circ \langle \mathbf{y} f_i \rangle_{i=1}^n)
\end{aligned}$$

as we wanted.

For multi-naturality:

$$\begin{aligned}
\beta_\pi \left\langle \Upsilon(\mathbf{y}[\pi_i] \xrightarrow{\alpha_i} F_i) \right\rangle_{i=1}^n &= \beta_\pi \left\langle \alpha_{i, \langle \langle \pi_{\langle i, j \rangle} \rangle_{k=1}^1}^{|\pi_i|} \langle \text{id}_{\pi_{\langle i, j \rangle}} \rangle_{j=1}^{|\pi_i|} \right\rangle_{i=1}^n \\
&\xrightarrow{\text{o-def}} (\beta \circ \langle \alpha_i \rangle_{i=1}^n)_\pi \langle \text{id}_{\pi_{\langle \ell \rangle - 1}} \rangle_{\ell=1}^{\|\pi\|} = \Upsilon(\beta \circ \langle \alpha_i \rangle_{i=1}^n)
\end{aligned}$$

as we wanted. ■

► **Lemma 14.** Special Yoneda reduction. *For every $\mathbf{a} \xrightarrow{u} b$, we have: $\Upsilon_{\mathcal{U}, \mathbf{y}b, \mathbf{a}}(\mathbf{y}[\mathbf{a}] \xrightarrow{\mathbf{y}u} \mathbf{y}b) = u$.*

Proof

Calculate: $\Upsilon(\mathbf{y}u) = \mathbf{y}u_{\mathbf{a}}(\text{id}_{\mathbf{a}_1}, \dots, \text{id}_{\mathbf{a}_n}) = u \circ (\text{id}, \dots, \text{id}) = u$. ■

► **Lemma 15.** *The Yoneda embedding is, up to equivalence, admissible: for every multi-category $\mathcal{U}$ there are:*

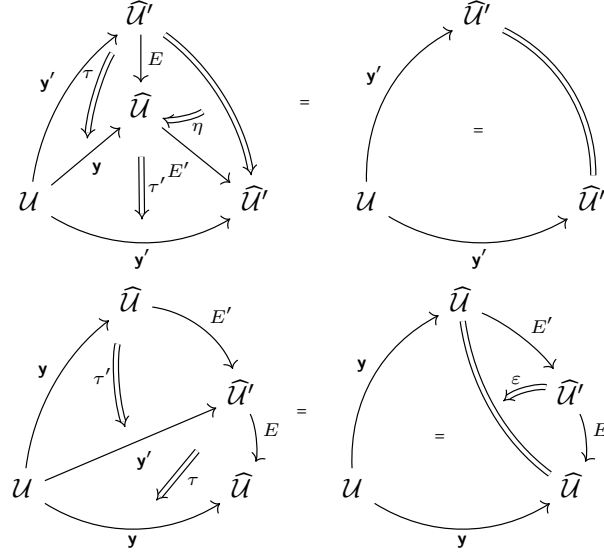
- a multi-category $\widehat{\mathcal{U}}'$;
- an admissible functor $\mathbf{y}'_{\mathcal{U}}: \mathcal{U} \rightarrow \widehat{\mathcal{U}}'$; and
- an adjoint equivalence in **MultiCAT**, i.e. an adjunction with invertible unit and counit:

$$\begin{array}{ccccc}
\widehat{\mathcal{U}}' & \xrightarrow{E} & \widehat{\mathcal{U}} & \xrightarrow{E'} & \widehat{\mathcal{U}}' & \xrightarrow{E} & \widehat{\mathcal{U}} \\
& \searrow \eta & & & \nearrow \varepsilon & & \\
& & & & & &
\end{array}$$

- 2-cells:

$$\begin{array}{ccc}
& \widehat{\mathcal{U}}' & \\
\mathbf{y}' \nearrow & & \searrow E \\
\mathcal{U} & \xrightarrow{\mathbf{y}} & \widehat{\mathcal{U}} \\
& \Downarrow \tau & \\
& &
\end{array}
\quad
\begin{array}{ccc}
& \widehat{\mathcal{U}} & \\
\mathbf{y} \nearrow & & \searrow E' \\
\mathcal{U} & \xrightarrow{\mathbf{y}'} & \widehat{\mathcal{U}}' \\
& \Downarrow \tau' & \\
& &
\end{array}$$

such that:

**Proof**

Take any $\mathbf{a}$ and presheaf F . By the Yoneda lemma:

$$\widehat{\mathcal{U}}(\mathbf{y}\mathbf{a}_1, \dots, \mathbf{y}\mathbf{a}_n; F) \cong F\mathbf{a} \in \mathbf{Set}$$

is a small set. So now we need to tediously construct a different presheaf multi-category $\widehat{\mathcal{U}}'$ which is admissible:

- $\text{Ob}(\widehat{\mathcal{U}}') := \text{Ob}(\widehat{\mathcal{U}}) + \text{Ob}(\mathcal{U})$, i.e., either a presheaf or an object, together with a tag telling us which is which. We call the left-tagged elements ‘actual presheaves’ and the right-tagged elements ‘formal representables’. For each $x \in \text{Ob}(\widehat{\mathcal{U}}')$, we can associate a presheaf $\llbracket x \rrbracket$:

$$\llbracket \iota_1 F \rrbracket := F \quad \llbracket \iota_2 a \rrbracket := \mathbf{y}a$$

- We define the homs as follows:

$$\widehat{\mathcal{U}}'(x_1, \dots, x_n; y) := \begin{cases} \forall i. x_i = \iota_2 a_i : \llbracket y \rrbracket(a_1, \dots, a_n) \\ \text{otherwise} : \widehat{\mathcal{U}}(\llbracket x_1 \rrbracket, \dots, \llbracket x_n \rrbracket; \llbracket y \rrbracket) \end{cases}$$

We can extend the semantics function into a multi-functor structure $\llbracket - \rrbracket : \widehat{\mathcal{U}}' \rightarrow \widehat{\mathcal{U}}$ using the Yoneda lemma:

$$\llbracket f \rrbracket_{x_1, \dots, x_n; y} := \begin{cases} \forall i. x_i = \iota_2 a_i : \Upsilon_{\widehat{\mathcal{U}}, \llbracket y \rrbracket, (a_1, \dots, a_n)}^{-1} f \\ \text{otherwise} : f \end{cases}$$

- The identities are the identity natural multi-transformations over actual presheaves, and the identity multi-morphism at a over the formal representable by a . By fiat, the semantics functor-structure preserves identities.
- To compose multi-morphisms, apply the semantics functor-structure to turn them into composable multi-natural transformations, compose them in $\widehat{\mathcal{U}}$, and then, if needed, apply the Yoneda lemma to get the corresponding element of the codomain presheaf. By fiat, the semantics functor-structure preserves composition.
- We therefore also have a functor from the multi-category structure $\widehat{\mathcal{U}}'$ to $\widehat{\mathcal{U}}$.

The functor $\llbracket - \rrbracket : \widehat{\mathcal{U}}' \rightarrow \widehat{\mathcal{U}}$ is fully-faithful, and in particular faithful, and therefore $\widehat{\mathcal{U}}'$ is a multi-category. We will take $E := \llbracket - \rrbracket$, and $E' := \iota_1 : \widehat{\mathcal{U}} \rightarrow \widehat{\mathcal{U}}'$.

We can define the unit and counit, since the component tells us whether we need a multi-transformation or merely its representation. For the unit we need to split into cases:

$$\eta_{\iota_1 F} := \text{id}_F \quad \eta_{\iota_2 a} := \text{id}_a \in \mathbf{y} a a = \llbracket \iota_1 \llbracket \iota_2 a \rrbracket \rrbracket a$$

For the counit, we know the domain is an actual presheaf, and so use the identity natural multi-transformation.

I expect these are all natural and satisfy the triangle laws.

The formal Yoneda functor $\mathbf{y}' : \mathcal{U} \rightarrow \widehat{\mathcal{U}}'$ sends each $a \in \text{Ob}(\mathcal{U})$ to its formal representable and each multi-morphism to itself. Since we started with a locally-small category, we have by fiat that $\mathbf{y}'$ is admissible:

$$\widehat{\mathcal{U}}'(\mathbf{y}' a_1, \dots, \mathbf{y}' a_n; x) = \llbracket x \rrbracket(a_1, \dots, a_n) \in \mathbf{Set}$$

The two transformations $\tau : \llbracket \mathbf{y}' - \rrbracket \rightarrow \mathbf{y}$ and $\tau' : \iota_1 \mathbf{y}' - \rightarrow \iota_1 -$ are then identities, since the hom definitions at each object take care of everything else.

What makes the two additional laws hold, in the case that we don't have an identity, is that the both the unit for the adjunction and our case-splitting composition will choose the correct representative. $\blacksquare$

5.5 The Restricted Yoneda multi-functor

The structure underlying a Yoneda structure is a choice of 4 components:

- The admissible Yoneda 1-cells $\mathbf{y}_{\mathcal{U}} : \mathcal{U} \rightarrow \widehat{\mathcal{U}}$, together with their presheaf construction. We covered both in the previous section, up to equivalence (Lemma 15).
- For every admissible functor $F : \mathcal{U} \rightarrow \mathcal{V}$, a 1-cell (non-admissible perhaps) $\mathcal{V}(F; -) : \mathcal{V} \rightarrow \widehat{\mathcal{U}}$;
- and a 2-cell $\text{map } F : \mathbf{y}_{\mathcal{U}} \rightarrow \mathcal{V}(F; F-)$.

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{F} & \mathcal{V} \\ & \searrow \text{map } F & \downarrow \mathcal{V}(F; -) \\ & \mathbf{y} & \widehat{\mathcal{U}} \end{array}$$

This section concerns constructing the remainder of this data, and proving the first axiom of the Yoneda structure.

Let $F : \mathcal{U} \rightarrow \mathcal{V}$ be an admissible multi-functor. To keep the notation less ellipsis-cluttered, given a sequence of objects/multi-morphisms χ in $\mathcal{U}$, we write $F[\chi] := \langle F\chi_i \rangle_{i=1}^{|\chi|}$ for the componentwise application of F .

We define the structure of the *F-restricted Yoneda multi-functor* $\mathcal{V}(F; -) : \mathcal{V} \rightarrow \widehat{\mathcal{U}}$:

- It sends each object $x \in \text{Ob}(\mathcal{V})$ to the homset functor $\mathcal{V}(F[-]; x) : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$ sending:
 - each sequence $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$ to the multi-homset $\mathcal{V}(F[\mathbf{a}]; x)$; and
 - each morphism $f : \mathbf{a} \leftarrow \mathbf{b}$ to the function:

$$\mathcal{V}(F[f]; x) : \left((F[\mathbf{a}]) \xrightarrow{u} x \right) \mapsto (F[\mathbf{b}]) \xrightarrow{u \circ (F[f])} x$$

- It sends each morphism $(\mathbf{x}) \xrightarrow{v} y$, $n := |\mathbf{x}|$ to the following natural multi-transformation $\mathcal{V}(F; v) : (\mathcal{V}(F; \mathbf{x}_1), \dots, \mathcal{V}(F; \mathbf{x}_n)) \rightarrow \mathcal{V}(F; y)$ given, for every n -ary partition π of $\mathbf{a}$, by:

$$\mathcal{V}(F; v)_\pi : \left\langle F[\pi_i] \xrightarrow{v_i} \mathbf{x}_i \right\rangle_{i=1}^n \mapsto (F[\mathbf{a}]) \xrightarrow{v \circ (v_1, \dots, v_n)} y$$

► **Lemma 16.** *These data define a multi-functor $\mathcal{V}(F; -) : \mathcal{V} \rightarrow \widehat{\mathcal{U}}$.*

Proof

The restricted Yoneda functor is well-defined:

- Each hom $\mathcal{V}(F[\mathbf{a}]; x)$ is a set since F is admissible.
- Given a morphism $\mathbf{a} \xleftarrow{f} \mathbf{b}$ in $\mathcal{U}$, $F[\mathbf{a}] \xleftarrow{F[f]} F[\mathbf{b}]$ in $\mathcal{V}$, and so given $F[\mathbf{a}] \xrightarrow{u} x$ the composition $u \circ (F[f]) : F[\mathbf{b}] \rightarrow x$ type-checks as a well-defined multi-morphism.
- For identity preservation, calculate:

$$\begin{array}{ccc} F \text{ multi-functor} & & \text{multi-category} \\ \downarrow & & \downarrow \\ \mathcal{V}(F[\text{id}_{\mathbf{a}}]; x)(u) := u \circ \langle F[\text{id}_{\mathbf{a}_i}] \rangle_{i=1}^{|\mathbf{a}|} & = & u \circ \langle \text{id}_{F[\mathbf{a}_i}]} \rangle_{i=1}^{|\mathbf{a}|} = u \end{array}$$

and so $\mathcal{V}(F[\text{id}_{\mathbf{a}}]; x) = \text{id}_{\mathcal{V}(F[\mathbf{a}]; x)}$.

- Take two composable morphisms in $\mathcal{U}^{\text{op}}$, i.e.: $\mathbf{a} \xleftarrow{f} \mathbf{b} \xleftarrow{g} \mathbf{c}$ in $\mathcal{U}$ and calculate for a generic $u : (F[\mathbf{a}]) \rightarrow x$:

$$\begin{array}{ccc} \circ\text{-def in } \mathcal{U}^{\text{op}} & & \\ \downarrow & & \\ \mathcal{V}(F[f \circ g]; x)u = u \circ (F[f \circ g]) & = & u \circ \left\langle F \left(f_i \circ \langle g_{(i,*)} \rangle_{j=1}^{|g_{(i,*)}|} \right) \right\rangle_{i=1}^{|f|} \\ & \begin{array}{ccc} F \text{ multi-functor} & & \text{associativity of composition} \\ & \downarrow & \text{in a multi-category} \end{array} \\ & = u \circ \left\langle F f_i \circ \langle F g_{(i,*)} \rangle_{j=1}^{|g_{(i,*)}|} \right\rangle_{i=1}^{|f|} & \downarrow \\ & = (u \circ \langle F f_i \rangle_{i=1}^{|f|}) \circ \langle F g_{\ell} \rangle_{\ell=1}^{|g|} & \\ & = \mathcal{V}(F[g]; x) \circ \mathcal{V}(F[f]; x)u & \end{array}$$

and so $\mathcal{V}(F[f \circ g]; x) = \mathcal{V}(F[g]; x) \circ \mathcal{V}(F[f]; x)$. Thus, $\mathcal{V}(F; x) : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Set}$ is a presheaf.

To show that the multi-functorial action is well-defined, take $(\mathbf{x}) \xrightarrow{v} y$ in $\mathcal{V}$ with $n := |\mathbf{x}|$, and show the naturality condition by taking an n -ary partition $\langle \pi_i \xleftarrow{\varphi_i} \pi'_i \rangle_{i=1}^n$ of some $\mathbf{a} \xleftarrow{f} \mathbf{b}$ in $\mathcal{U}$, chasing an element of $\prod_{i=1}^n \mathcal{V}(F[\mathbf{a}]; \mathbf{x}_i)$:

$$\begin{array}{ccc} \left\langle (F[\pi_i]) \xrightarrow{u_i} \mathbf{x}_i \right\rangle_{i=1}^n & \xrightarrow{\mathcal{V}(F; v)_\pi} & v \circ \langle u_i \rangle_{i=1}^n \\ \downarrow \prod_{i=1}^n \mathcal{V}(F[\varphi_i]; \mathbf{x}_i) & & \downarrow \mathcal{V}(F[f]; y) \\ \left\langle (F[\pi'_i]) \xrightarrow{u_i \circ F[\varphi_{(i,*)}]} \mathbf{x}_i \right\rangle_{i=1}^n & \xrightarrow{\mathcal{V}(F; v)_{\pi'}} & v \circ \langle u_i \circ (F[\varphi_{(i,*)}]) \rangle_{i=1}^n \\ & \begin{array}{ccc} \varphi \text{ partitions } f \rightarrow & & \\ \uparrow & & \text{associative composition} \\ & & \text{in a multi-category} \end{array} & \\ & & (v \circ \langle u_i \rangle_{i=1}^n) \circ F[f] & \\ & & \uparrow & \\ & & (v \circ \langle u_i \rangle_{i=1}^n) \circ F[\varphi_{(*, \#)}] & \end{array}$$

To show this action is multi-functorial, take an identity id_x in $\mathcal{V}$ and the unique unary partition $\langle \mathbf{a} \rangle$ of some $\mathbf{a}$:

$$\begin{array}{ccc} \langle F[\mathbf{a}] \xrightarrow{u} x \rangle_{i=1}^1 & \xrightarrow{\mathcal{V}(F; \text{id}_x)_{\langle \mathbf{a} \rangle}} & \text{id}_x \circ \langle u \rangle_{i=1}^n \\ & \searrow \text{id}_{\mathcal{V}(F; \text{id}_x)} \cong & \downarrow \text{←neutrality} \\ & & u \end{array}$$

Next, given composable multi-morphisms $(\mathbf{y}) \xrightarrow{v} z$, $n := |\mathbf{y}|$ and $\left\langle (\mathbf{x}_i) \xrightarrow{v_i} \mathbf{y}_i \right\rangle_{i=1}^n$ in $\mathcal{V}$, letting $m := \|\mathbf{x}\|$ and $m_i := |\mathbf{x}_i|$ and taking any m -ary partition of some $\mathbf{a}$, calculate:

$$\begin{array}{c}
\langle F[\pi_\ell] \rangle_{\ell=1}^m \xrightarrow{u_\ell} \mathbf{x}_\ell \xrightarrow{\cong} \langle \langle u_{\langle i,j \rangle} \rangle_{j=1}^{m_i} \rangle_{i=1}^n \xrightarrow{\prod_{i=1}^n \mathcal{V}(F; v_i)_{\pi_{\langle i, * \rangle}}} \langle v_i \circ \langle u_{\langle i,j \rangle} \rangle_{j=1}^{m_i} \rangle_{i=1}^n \\
\downarrow \text{circ-def} = \downarrow \mathcal{V}(F; v)_{\pi_{\langle i, * \rangle}} \quad \downarrow \leftarrow \text{associativity} \\
\langle F[\pi_\ell] \rangle_{\ell=1}^m \xrightarrow{\mathcal{V}(F; v)_{\pi_{\langle i, * \rangle}}} v \circ \langle v_i \circ \langle u_{\langle i,j \rangle} \rangle_{j=1}^{m_i} \rangle_{i=1}^n
\end{array}$$

So $\mathcal{V}(F; v \circ \langle v_i \rangle_{i=1}^n) = \mathcal{V}(F; v) \circ \langle \mathcal{V}(F; v_i) \rangle_{i=1}^n$ and we have a functor,

The multi-natural transformation $\mathbf{map} F : \mathbf{y}_{\mathcal{U}} \Rightarrow \mathcal{V}(F; F-)$ we now define internalises the multi-functorial action of F . Explicitly, the component at $b \in \text{Ob}(\mathcal{U})$ is given by the natural multi-transformation $(\mathbf{map} F)_b : (\mathbf{y}_{\mathcal{U}} b) \rightarrow \mathcal{V}(F; Fb)$ whose component at any unary partition $\langle \mathbf{a} \rangle_{i=1}^1$ of some sequence $\mathbf{a}$ is given by:

$$(\text{map } F)_{b, \langle \mathbf{a} \rangle_{i=1}^1} : \prod \{ \mathbf{y} \mathbf{b} \mathbf{a} \} \ni \left\langle (\mathbf{a} \xrightarrow{u} b) \right\rangle_{i=1}^1 \mapsto (F[\mathbf{a}] \xrightarrow{Fu} Fb) \in \mathcal{V}(F[\mathbf{a}]; Fb)$$

or, equivalently, using the Yoneda lemma:

$$(\text{map } F)_{b, \langle \mathbf{a} \rangle_{i=1}^1} \langle u \rangle := \Upsilon_{F; \mathbf{a}}(\mathbf{y}(F[\mathbf{a}])) \xrightarrow{\mathbf{y}(Fu)} \mathbf{y}(Fb) \in \mathbf{y}(Fb)(F[\mathbf{a}]) = \mathcal{V}(F[\mathbf{a}]; Fb)$$

► **Lemma 17.** *This data defines a multi-natural transformation: $\text{map } F : \mathbf{y}_{\mathcal{U}} \Rightarrow \mathcal{V}(F; F-)$.*

Proof

By Yoneda reduction, the two definitions are equivalent. We need to show:

- Well-definedness: each component is multi-natural.
- Naturality of the components over-all.

Unfortunately, these don't just follow automatically from our formulation using the Yoneda lemma together with its naturality and multi-naturality, since we're applying F to u . However, the formulation using Yoneda might give some intuition why this operation is multi-natural, and our later results will allow us to recover it internally.

To show well-definedness, we need to show that each component $(\mathbf{map} F)_b$ is a natural multi-transformation. Take the unique unary partition $\langle f \rangle_{i=1}^1$ of some morphism $f : \mathbf{a} \leftarrow \mathbf{a}'$ in $\mathcal{U}$, and chase an element of $\prod \{\mathbf{yba}\}$ around the naturality square:

$$\begin{array}{ccc}
\langle (a \xrightarrow{u} b) \rangle_{i=1}^1 & \xrightarrow{(\text{map } F)_{b, \langle a \rangle}} & (F[a] \xrightarrow{Fu} Fb) \\
\downarrow \prod \{ybf\} & & \downarrow \mathcal{V}(F[f]; Fb) \\
\langle u \circ (f) \rangle_{i=1}^1 & \xrightarrow{\mathcal{V}(\text{map } F)_{b, \langle a' \rangle_{i=1}^1}} & F(u \circ (f))
\end{array}$$

multi-functoriality $\rightarrow$ $\parallel$

To show multi-naturality, take any $f : \mathbf{b} \rightarrow b$ in $\mathcal{U}$, let $n := |\mathbf{b}|$ and consider one component of the two sides of the multi-naturality diagram in $\widehat{\mathcal{U}}$:

$$(\text{map } F)_b \circ (\mathbf{y}f) = \mathcal{V}(F; Ff) \circ \langle (\text{map } F)_{\mathbf{b}_i} \rangle_{i=1}^n$$

by taking an n -ary partition π of some $\mathbf{a}$ and calculating:

$$\begin{array}{ccc}
\langle \pi_\ell \xrightarrow{u_\ell} \mathbf{b}_\ell \rangle_{\ell=1}^n & \xrightarrow{(\mathcal{V}(F; Ff) \circ \langle (\text{map } F)_{\mathbf{b}_i} \rangle_{i=1}^n)_\pi} & \\
\swarrow \cong & \searrow \cong & \\
\langle \langle (\pi_i) \xrightarrow{u_i} \mathbf{b}_i \rangle_{j=1}^1 \rangle_{i=1}^n & \xrightarrow{\prod_{i=1}^n (\text{map } F)_{\mathbf{b}_i, \pi(i, *)}} & \langle F[\pi_i] \xrightarrow{Fu_i} F\mathbf{b}_i \rangle_{i=1}^n \\
\downarrow \prod \{(\mathbf{y}f)_\pi\} & & \downarrow \mathcal{V}(F; Ff)_\pi \\
\langle f \circ \langle u_i \rangle_{i=1}^n \rangle_{k=1}^1 & \xrightarrow{(\text{map } F)_{b, \pi}} & Ff \circ \langle Fu_i \rangle_{i=1}^n \\
\downarrow \text{o-def} = & & \downarrow \text{multi-functoriality} \\
((\text{map } F)_b \circ (\mathbf{y}f))_\pi & \xrightarrow{\quad} & F(f \circ \langle u_i \rangle_{i=1}^n)
\end{array}$$

and so $\text{map } F$ is multi-natural. ■

► **Proposition 18.** Axiom 1. The multi-functor $\mathcal{V}(F; -)$ together with the multi-natural transformation $\text{map } F$ form the left Kan extension of $\mathbf{y}_{\mathcal{U}}$ along F :

$$\begin{array}{ccc}
\mathcal{U} & \xrightarrow{F} & \mathcal{V} \\
& \searrow \text{map } F & \downarrow \mathcal{V}(F; -) \\
& & \widehat{\mathcal{U}}
\end{array}$$

Proof

Like the Yoneda lemma, we roll up our sleeves and just do it. We need to show a bijective correspondence:

$$\begin{array}{ccc}
\mathcal{U} & \xrightarrow{F} & \mathcal{V} \\
\searrow \text{y}_{\mathcal{U}} & \nearrow \alpha & \downarrow \beta \\
& & \widehat{\mathcal{U}}
\end{array}
\quad \xleftrightarrow[\text{map } F]{\text{pasting}} \quad
\mathcal{V}(F; -) \left(\begin{array}{ccc} \mathcal{V} & & \\ \beta & \xrightarrow{\quad} & \\ \widehat{\mathcal{U}} & & \end{array} \right)_G$$

Starting with uniqueness, let $\alpha := (\beta \bullet F) \circ (\text{map } F)$ be the pasted multi-natural transformation.

First, unpack the determining structure of β . Take any component $x \in \text{Ob } (\mathcal{U})[2]$ of the multi-natural transformation $\beta : \mathcal{V}(F; -) \Rightarrow G$, giving a natural multi-transformation $\beta_x : (\mathcal{V}(F; x)) \rightarrow Gx$. Take any component of β_x , i.e. a unary partition of some $\mathbf{a} \in \text{Ob } (\mathcal{U}^{\text{op}})$ and we now have:

$$\beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} : \mathcal{V}(F[\mathbf{a}]; x) \rightarrow Gx(\mathbf{a})$$

and we need to determine the image of $\beta_{x, \langle \mathbf{a} \rangle_{k=1}^1}$ on some $F[\mathbf{a}] \xrightarrow{v} x$, and show that it only depends on α .

Note that we have a natural multi-transformation $Gv : (GF[\mathbf{a}]) \rightarrow Gx$ in $\widehat{\mathcal{U}}$ and so we may use the naturality of each component of α , form the composite and apply the Yoneda lemma to it:

$$Gv \circ \left\langle \mathbf{y}_{\mathbf{a}_i} \xrightarrow{\alpha_{\mathbf{a}_i}} GF\mathbf{a}_i \right\rangle_{i=1}^n : (\mathbf{y}[\mathbf{a}]) \rightarrow Gx \implies \Upsilon_{Gx, \mathbf{a}}(Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^n) \in Gx(\mathbf{a})$$

We will show that:

$$\beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle v \rangle_{k=1}^1 = \Upsilon_{Gx, \mathbf{a}}(Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^n)$$

and since the latter only involves α , and not β , then pasting with $\text{map } F$ is injective. Indeed:

$$\begin{aligned} \Upsilon_{Gx, \mathbf{a}}(Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^n) &= (Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^n) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \text{id}_{\mathbf{a}_i} \rangle_{i=1}^n \\ &= (Gv) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \alpha_{\mathbf{a}_i}(\text{id}_{\mathbf{a}_i}) \rangle_{i=1}^n \\ \alpha &= (\beta \bullet F) \circ (\text{map } F) \\ &\downarrow \\ &= (Gv) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \beta_{F\mathbf{a}_i}(F\text{id}_{\mathbf{a}_i}) \rangle_{i=1}^n \\ \text{multi-functoriality} &\downarrow \\ &= (Gv) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \beta_{F\mathbf{a}_i}(\text{id}_{F\mathbf{a}_i}) \rangle_{i=1}^n \\ \text{o-def in } \widehat{\mathcal{U}} &\downarrow \\ &= (Gv \circ \langle \beta_{F\mathbf{a}_i} \rangle_{i=1}^n) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \text{id}_{F\mathbf{a}_i} \rangle_{i=1}^n \\ \text{multi-naturality} &\downarrow \\ &= (\beta_x \circ (\mathcal{V}(F; v))) \langle \langle \mathbf{a}_i \rangle_{j=1}^1 \rangle_{i=1}^n \langle \text{id}_{F\mathbf{a}_i} \rangle_{i=1}^n \\ \text{o-def} &\downarrow \\ &= \beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle \mathcal{V}(F; v) \rangle_{\langle \mathbf{a}_i \rangle_{i=1}^n} \langle \text{id}_{F\mathbf{a}_i} \rangle_{i=1}^n \langle \rangle_{k=1}^1 \\ \mathcal{V}(F; v)\text{-def} \quad \text{multi-category} &\downarrow \quad \downarrow \\ &= \beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle v \circ \langle \text{id}_{F\mathbf{a}_i} \rangle_{i=1}^n \rangle_{k=1}^1 = \beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle v \rangle_{k=1}^1 \end{aligned}$$

and so the correspondence is injective.

To show surjectivity, consider any multi-natural α , and set:

$$\beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle v \rangle_{k=1}^1 := \Upsilon_{Gx, \mathbf{a}}(Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^n)$$

We need to show:

- β_x is a natural multi-transformation;
- β is a multi-natural transformation;
- pasting with $\text{map } F$ recovers α , i.e.: $\alpha = (\beta \bullet F) \circ (\text{map } F)$.

To show that $\beta_x : \mathcal{V}(F; x) \rightarrow Gx$ is natural, take any $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$ and calculate:

$$\begin{aligned}
 Gxf(\beta_{x, \langle \mathbf{a} \rangle_{k=1}^1} \langle v \rangle_{k=1}^1) &= Gxf \left(\Upsilon(Gv \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^{|\mathbf{a}|}) \right) \stackrel{\Upsilon\text{-naturality}}{=} \Upsilon \left(Gv \circ \langle \alpha_{\mathbf{a}_i} \circ \mathbf{y}f_i \rangle_{i=1}^{|\mathbf{a}|} \right) \\
 &\stackrel{\alpha\text{-multi-naturality}}{=} \Upsilon \left(Gv \circ \left\langle GFf_i \circ \langle \alpha_{\mathbf{b}_{(i,j)}} \rangle_{j=1}^{|\mathbf{b}_{(i,*)}|} \right\rangle_{i=1}^{|\mathbf{a}|} \right) \\
 &\stackrel{G\text{-multi-functor}}{=} \Upsilon \left(G(v \circ Ff_i) \circ \langle \alpha_{\mathbf{b}_\ell} \rangle_{\ell=1}^{|\mathbf{b}|} \right) = \beta_{x, \langle \mathbf{b} \rangle_{k=1}^1} \langle \mathcal{V}(F[f]; x)v \rangle_{k=1}^1
 \end{aligned}$$

To show that β is multi-natural, take any $v : (\mathbf{x}) \rightarrow y$ in $\mathcal{V}$ and $v_i : F[\mathbf{a}_i] \rightarrow \mathbf{x}_i$ for $1 \leq i \leq n := |\mathbf{a}|$:

$$\begin{aligned}
 (Gv \circ \langle \beta_{\mathbf{x}_i} \rangle_{i=1}^n)_{\langle \mathbf{a}_i \rangle_{i=1}^n} (v_1, \dots, v_n) &= Gv \left\langle \Upsilon(Gv_i \circ \langle \alpha_{\mathbf{a}_{(i,j)}} \rangle_{j=1}^{|\mathbf{a}_i|}) \right\rangle_{i=1}^n \\
 &\stackrel{\Upsilon\text{-multi-naturality, associativity}}{=} \Upsilon((Gv \circ \langle Gv_i \rangle_{i=1}^n) \circ \langle \alpha_{\mathbf{a}_\ell} \rangle_{\ell=1}^{|\mathbf{a}|}) \\
 &\stackrel{G\text{-multi-functoriality}}{=} \Upsilon(G(v \circ \langle v_i \rangle_{i=1}^n) \circ \langle \alpha_{\mathbf{a}_\ell} \rangle_{\ell=1}^{|\mathbf{a}|}) \\
 &= (\beta_y \circ \mathcal{V}(F; v)) \langle v_1, \dots, v_n \rangle_{\ell=1}^n
 \end{aligned}$$

To show that we recover α , take any $u : (\mathbf{a}) \rightarrow b$ and calculate:

$$((\beta \bullet F) \circ (\text{map } F))_{b, \langle \mathbf{a} \rangle_{k=1}^1} (u) = \beta(Fu) = \Upsilon(GFu \circ \langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^{|\mathbf{a}|}) \stackrel{\alpha\text{-multi-naturality}}{=} \Upsilon(\alpha_b \circ \mathbf{y}u) = \alpha_{b, \langle \mathbf{a} \rangle_{k=1}^1} u$$

concluding the proof. ■

5.6 The Relative Yoneda lemma

► **Proposition 19.** *Axiom 2. For every multi-functor (admissible or not) $G : \mathcal{W} \rightarrow \mathcal{U}$, the following triangle is a left Kan lifting of $\mathbf{y}_{\mathcal{U}}$ through $\mathcal{V}(F; -)$:*

$$\begin{array}{ccccc}
 \mathcal{W} & \xrightarrow{G} & \mathcal{U} & \xrightarrow{F} & \mathcal{V} \\
 & \searrow G & \parallel & \nearrow \text{map } F & \downarrow \mathcal{V}(F; -) \\
 & & \mathcal{U} & & \\
 & & \searrow \mathbf{y}_{\mathcal{U}} & & \\
 & & & & \hat{\mathcal{U}}
 \end{array}$$

I.e., for every $H : \mathcal{W} \rightarrow \mathcal{V}$, pasting $(\text{map } F) \bullet G$ along $\mathcal{W} \xrightarrow{G} \mathcal{U} \xrightarrow{F} \mathcal{V}$ gives a bijection:

$$\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \alpha & \downarrow \mathcal{V}(F; -) \\
& \mathcal{U} & \\
& \searrow y_{\mathcal{U}} & \\
& \widehat{\mathcal{U}} &
\end{array}
\longleftrightarrow
\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \Upsilon_{\mathcal{U}, F, H, G(-)} \alpha & \downarrow F \\
& \mathcal{U} &
\end{array}$$

We call the bijection $\alpha \mapsto \Upsilon_{\mathcal{U}, F, H, G(-)} \alpha$ the Yoneda lemma relative to G along F .

Proof

Starting with uniqueness, given the data:

$$\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \alpha & \downarrow \mathcal{V}(F; -) \\
& \mathcal{U} & \\
& \searrow y_{\mathcal{U}} & \\
& \widehat{\mathcal{U}} &
\end{array}
\quad
\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \beta & \downarrow F \\
& \mathcal{U} &
\end{array}$$

such that

$$\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \beta & \downarrow F \\
& \mathcal{U} & \\
& \searrow y_{\mathcal{U}} & \\
& \widehat{\mathcal{U}} &
\end{array}
=
\begin{array}{ccc}
\mathcal{W} & \xrightarrow{H} & \mathcal{V} \\
\downarrow G & \nearrow \alpha & \downarrow \mathcal{V}(F; -) \\
& \mathcal{U} & \\
& \searrow y_{\mathcal{U}} & \\
& \widehat{\mathcal{U}} &
\end{array}$$

we will show that β is uniquely determined by α , i.e., the relative Yoneda lemma is injective.

Take any $e \in \text{Ob}(\mathcal{W})$, and then: $\beta_e : (FGe) \rightarrow He$ is a multi-morphism in $\mathcal{V}$, i.e. an element $\beta_e \in \mathcal{V}(F[(Ge)]; He)$, and so by the Yoneda lemma we have a multi-natural transformation:

$$\alpha'_e := \Upsilon_{\mathcal{U}, \mathcal{V}(F; He), (Ge)}^{-1} \beta_e : \mathbf{y}(Ge) \rightarrow \mathcal{V}(F[-]; He)$$

which is the same type as the component α_e . We will show that $\alpha'_e = \alpha_e$, and so β_e is uniquely determined by α :

$$\beta_e = \Upsilon_{\mathcal{U}, \mathcal{V}(F; He), (Ge)} \alpha_e$$

Take any $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$ and $u : (\mathbf{a}) \rightarrow Ge$ in $\mathcal{U}$. On the one hand:

$$\alpha'_{e, \mathbf{a}}(u) \stackrel{\Upsilon\text{-def}}{\downarrow} \mathcal{V}(F[u]; He)(\beta_e) = \beta_e \circ (Fu)$$

On the other hand:

$$\alpha_{e, \mathbf{a}}(u) \stackrel{\text{assumption}}{\downarrow} ((\mathcal{V}(F; -) \bullet \beta) \circ (\text{map } F \bullet G))_{e, \mathbf{a}}(u) \stackrel{(\bullet\text{-def}) \circ (\text{map } F\text{-def})}{\downarrow} \mathcal{V}(F; \beta_e)(Fu) = \beta_e \circ (Fu)$$

and therefore β is uniquely determined by α .

For the existence, set $\beta_e := \Upsilon_{\mathcal{U}, \mathcal{V}(F; He), (Ge)} \alpha_e$ — this choice motivates our notation — and we'll show:

- $\beta : FG \rightarrow H$ is multi-natural; and
- $\alpha = (\mathcal{V}(F; -) \bullet \beta) \circ (\text{map } F \bullet G)$

For multi-naturality, take any $(e_1, \dots, e_n) \xrightarrow{w} e$ in $\mathcal{W}$. We want to prove:

$$Hw \circ (\beta_{e_1}, \dots, \beta_{e_n}) = \beta_e \circ (FGw)$$

Calculate:

$$\begin{aligned} Hw \circ (\beta_{e_1}, \dots, \beta_{e_n}) &= \mathcal{V}(F; Hw) \langle \Upsilon(\alpha_{e_i}) \rangle_{i=1}^n \xrightarrow{\Upsilon\text{-multi-naturality}} \Upsilon(\mathcal{V}(F; Hw) \circ \langle \alpha_{e_i} \rangle_{i=1}^n) \\ &= \Upsilon(\alpha_e \circ (\mathbf{y}(Gw))) = \mathcal{V}(F[Gw]; He)(\Upsilon(\alpha_e)) = \beta_e \circ (FGw) \\ &\quad \uparrow \quad \quad \quad \uparrow \\ &\quad \alpha : \mathbf{y}(G-) \rightarrow \mathcal{V}(F; H-) \quad \Upsilon\text{-naturality} \\ &\quad \text{multi-naturality} \end{aligned}$$

as we wanted.

To show that we recover α by pasting, take any $e \in \mathcal{W}$, $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$, $u : (\mathbf{a}) \rightarrow Ge$. We need to show:

$$\alpha_{e, \mathbf{a}}(u) = \mathcal{V}(F; \beta_e)(\text{map } Fu) = \beta_e \circ (Fu)$$

Calculate:

$$\beta_e \circ (Fu) = \mathcal{V}(F[u]; He)(\Upsilon(\alpha_e)) \xrightarrow{\Upsilon\text{-naturality}} \Upsilon(\alpha_e \circ (\mathbf{y}u)) \xrightarrow{\Upsilon\text{-multi-naturality}} \alpha_{e, \mathbf{a}}(\Upsilon(\mathbf{y}u)) \xrightarrow{\text{Yoneda reduction}} \alpha_{e, \mathbf{a}}u$$

as we wanted. ■

5.7 The presheaf-construction pseudo-functor

The third axioms concerns internalising the presheaf construction into a pseudo-functor from the locally-small multi-categories to multi-categories. Doing so uses an internalisation of the hom profunctor that extends the postulate structure $\mathcal{V}(F; -)$, first to an action on F , and then to a similar action in the covariant position.

Internalising the hom profunctor. The following structure and properties follow from Axiom 1, letting $\mathbf{MultiCAT}_{\text{admissible}}$ denote that full sub-2-category of $\mathbf{MultiCAT}_{\text{admissible}}$ consisting of the admissible multi-functors, we have, for every triple of multi-categories $\mathcal{U}, \mathcal{V}, \mathcal{W}$, with $\mathcal{U}$ locally-small, a functor:

$$\mathcal{V}(-, --) : \mathbf{MultiCAT}_{\text{admissible}}(\mathcal{U}, \mathcal{V})^{\text{op}} \times \mathbf{MultiCAT}(\mathcal{W}, \mathcal{V}) \rightarrow \mathbf{MultiCAT}(\mathcal{W}, \widehat{\mathcal{U}})$$

To define it, we first define its partial application with $\mathcal{W} := \mathcal{V}$, $\text{Id} : \mathcal{W} \rightarrow \mathcal{V}$, namely:

$$\mathcal{V}(-, \text{Id}-) : \mathbf{MultiCAT}_{\text{admissible}}(\mathcal{U}, \mathcal{V})^{\text{op}} \rightarrow \mathbf{MultiCAT}(\mathcal{V}, \widehat{\mathcal{U}})$$

given by:

- sending ‘objects’, i.e., admissible multi-functors $F : \mathcal{U} \rightarrow \mathcal{V}$, to the postulated multi-functor:

$$\mathcal{V}(F, \text{Id}-) := \mathcal{V}(F; -) : \mathcal{V} \rightarrow \widehat{\mathcal{U}}$$

- sending ‘morphisms’, i.e., multi-natural transformations $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$, to the unique multi-natural transformation:

$$\mathcal{V} \begin{array}{c} \xrightarrow{\mathcal{V}(F; \text{Id}-)} \\ \Downarrow \mathcal{V}(\alpha; \text{Id}-) \\ \xrightarrow{\mathcal{V}(G; \text{Id}-)} \end{array} \widehat{\mathcal{U}}$$

satisfying:

$$\begin{array}{c} \mathcal{U} \xrightarrow{F} \mathcal{V} \\ \text{map } F \downarrow \mathcal{V}(F; \text{Id}-) \\ \mathbf{y} \searrow \mathcal{V}(\alpha; \text{Id}-) \searrow \mathcal{V}(G; \text{Id}-) \\ \widehat{\mathcal{U}} \end{array} = \begin{array}{c} \mathcal{U} \xrightarrow{F} \mathcal{V} \\ \text{map } G \downarrow \mathcal{V}(G; \text{Id}-) \\ \mathbf{y} \searrow \mathcal{V}(\alpha; \text{Id}-) \searrow \mathcal{V}(G; \text{Id}-) \\ \widehat{\mathcal{U}} \end{array}$$

Working this action out explicitly, it's given by the multi-natural transformation consisting of natural multi-transformations acting by pre-composition with the appropriate component of α :

$$\mathcal{V}(\alpha; \text{Id}-)_{x, \mathbf{a}} : \left(F[\mathbf{a}] \xrightarrow{u} x \right) \mapsto \left(G[\mathbf{a}] \xrightarrow{\langle \alpha_{\mathbf{a}} \rangle_i} F[\mathbf{a}] \xrightarrow{u} \right)$$

We then define the more general functor using horizontal composition as follows. Given:

$$\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V} \begin{array}{c} \xleftarrow{H} \\ \Downarrow \gamma \\ \xleftarrow{K} \end{array} \mathcal{W}$$

we define:

$$\mathcal{W} \begin{array}{c} \xrightarrow{\mathcal{V}(F; H-)} \\ \Downarrow \mathcal{V}(\alpha; \gamma) \\ \xrightarrow{\mathcal{V}(G; K-)} \end{array} \widehat{\mathcal{U}} \quad := \quad \mathcal{W} \begin{array}{c} \xrightarrow{H} \\ \Downarrow \gamma \\ \xrightarrow{K} \end{array} \mathcal{V} \begin{array}{c} \xrightarrow{\mathcal{V}(F; \text{Id}-)} \\ \Downarrow \alpha \\ \xrightarrow{\mathcal{V}(G; \text{Id}-)} \end{array} \widehat{\mathcal{U}}$$

Explicitly, the functor $\mathcal{V}(-; --)$ is given by:

- sending ‘objects’, i.e., admissible multi-functors $F : \mathcal{U} \rightarrow \mathcal{V}$ and arbitrary multifunctor $H : \mathcal{W} \rightarrow \mathcal{V}$, to the composed multi-functor:

$$\mathcal{V}(F, H-) := \mathcal{V}(F; -) \circ H : \mathcal{W} \rightarrow \mathcal{V} \rightarrow \widehat{\mathcal{U}}$$

Its object function is given by:

- for all $x \in \text{Ob}(\mathcal{V})$ and $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$:

$$\mathcal{V}(F; H-)x \mathbf{a} := \mathcal{V}(F[\mathbf{a}]; Hx)$$

- for all $x \in \text{Ob}(\mathcal{V})$ and $\mathbf{a} \xleftarrow{f} \mathbf{b}$ in $\mathcal{U}$:

$$\mathcal{V}(F; H-)x f := \mathcal{V}(F[f]; Hx) : ((F[\mathbf{a}]) \xrightarrow{u} Hx) \mapsto ((F[\mathbf{b}]) \xrightarrow{F[f]} (F[\mathbf{a}]) \xrightarrow{u} Hx)$$

Its hom action is given by:

- for all $g : \mathbf{x} \rightarrow y$ in $\mathcal{V}$ and $\mathbf{a} \in \text{Ob}(\mathcal{U}^{\text{op}})$:

$$\mathcal{V}(F; Hg)_{\mathbf{a}} := \mathcal{V}(F[\mathbf{a}]; Hg) : \left\langle (F[\mathbf{a}]) \xrightarrow{u_i} Hx \right\rangle_{i=1}^{|\mathbf{x}|} \mapsto \left(F[\mathbf{a}] \xrightarrow{Hg \circ \langle u_i \rangle_i} Hy \right)$$

- and its operation on ‘morphisms’:

$$\begin{array}{ccc} \mathcal{U} & \begin{array}{c} \xrightarrow{F} \\ \uparrow \alpha \\ \xrightarrow{G} \end{array} & \mathcal{V} \\ & \begin{array}{c} \xleftarrow{H} \\ \downarrow \gamma \\ \xleftarrow{K} \end{array} & \mathcal{W} \end{array} \xrightarrow{\mathcal{V}(-; --)} \begin{array}{ccc} & \xrightarrow{\mathcal{V}(F; H-)} & \\ & \downarrow \mathcal{V}(\alpha; \gamma) & \\ & \xrightarrow{\mathcal{V}(G; K-)} & \end{array} \mathcal{U}$$

is given by the multi-natural transformation whose component at x is the natural multi-transformation whose component at $\mathbf{a}$ is:

$$\mathcal{V}(\alpha; \gamma)_{x, \mathbf{a}} : \left((F[\mathbf{a}]) \xrightarrow{u} Hx \right) \mapsto \left((G[\mathbf{a}]) \xrightarrow{\langle \alpha_{\mathbf{a}_i} \rangle_{i=1}^{|\mathbf{a}|}} (F[\mathbf{a}]) \xrightarrow{u} Hx \xrightarrow{\gamma_x} Kx \right)$$

Street and Walters then use this functor, still only with Axiom 1, to extend the presheaf construction to a co-lax (2-)functor from the 2-category of locally-small multi-categories to multi-categories $\widehat{\cdot} : \mathbf{MultiCAT}_{\mathcal{L}}^{\text{coop}} \rightarrow \mathbf{MultiCAT}$, sending:

- each multi-category $\mathcal{U}$ to its presheaf construction $\widehat{\mathcal{U}}$;
- each (necessarily admissible) multi-functor $F : \mathcal{U} \leftarrow \mathcal{V}$ between locally-small categories to the multi-functor: $\widehat{F} := \widehat{\mathcal{U}}(\mathcal{U}(\text{Id}; F-), -) : \widehat{\mathcal{U}} \rightarrow \widehat{\mathcal{V}}$ given explicitly by:
 - sending each presheaf $P \in \widehat{\mathcal{U}}$ to the presheaf $\widehat{F}P := \widehat{\mathcal{U}}(\mathcal{U}(\text{Id}; F-), P) \in \widehat{\mathcal{V}}$:
 - * sending each $\mathbf{x} \in \mathcal{V}^{\text{op}}$ to the set of $|\mathbf{x}|$ -ary natural multi-transformations of type $\beta : (\mathcal{U}(\text{Id}; F[\mathbf{x}])) \rightarrow P$, i.e. $\beta : \mathbf{y}_{\mathcal{U}}(F[\mathbf{x}]) \rightarrow P$;
 - * sending each $\mathbf{x} \xleftarrow{g} \mathbf{y}$ in $\mathcal{V}$ to the natural multi-transformation that acts by pre-composing with the multi-transformation that acts by post-composing Fg :

$$\begin{aligned} \widehat{F}g : \left(\mathbf{y}_{\mathcal{U}}(F[\mathbf{x}]) \xrightarrow{\beta} P \right) &\mapsto \left(\mathbf{y}_{\mathcal{U}}(F[\mathbf{y}]) \xrightarrow{\mathbf{y}_{\mathcal{U}}(F[g])} \mathbf{y}_{\mathcal{U}}(F[\mathbf{x}]) \xrightarrow{\beta} P \right) \\ &= \left\langle \left((\pi_{\ell}) \xrightarrow{u_{\ell}} F\mathbf{y}_{\ell} \right)_{\ell=1}^{|\mathbf{y}|} \mapsto \beta_{\pi(\ast, \ast)} \circ (Fg_i \circ \langle u_{\langle i, j \rangle} \rangle_{j=1}^{|\mathbf{y}_{\langle i, \ast \rangle}|}) \right\rangle_{\pi} \end{aligned}$$

- sending each natural multi-transformation $\alpha : (\mathbf{P}) \rightarrow Q$ in $\widehat{\mathcal{U}}$ to the natural multi-transformation $\widehat{F}\alpha : (\widehat{F}[\mathbf{P}]) \rightarrow \widehat{F}Q$ in $\widehat{\mathcal{V}}$ given at partition π of $\mathbf{x} \in \text{Ob}(\mathcal{V})$ by:

$$\left\langle \mathbf{y}(F[\pi_i]) \xrightarrow{\beta_i} P_i \right\rangle_{i=1}^{|\mathbf{P}|} \mapsto \left\langle \mathbf{y}(F\mathbf{x}_{\ell}) \right\rangle_{\ell=1}^{|\mathbf{x}|} \xrightarrow{\alpha_{F[[\pi]]} \circ \left\langle \beta_{i, \langle \pi_i \rangle_{j=1}^1} \right\rangle_{i=1}^{|\mathbf{P}|}} Q\mathbf{x}$$

- each multi-natural transformation $\mathcal{U} \begin{array}{c} \xleftarrow{F} \\ \uparrow \alpha \\ \xleftarrow{G} \end{array} \mathcal{V}$ to the multi-natural transformation

whose $\widehat{\mathcal{U}} \begin{array}{c} \xleftarrow{\widehat{F}} \\ \downarrow \widehat{\alpha} \\ \xleftarrow{\widehat{G}} \end{array} \widehat{\mathcal{V}}$ component at $P \in \widehat{\mathcal{U}}$ is the unary natural multi-transformation whose

component at a partition $\langle \mathbf{x} \rangle_{j=1}^1$ of some $\mathbf{x} \in \mathcal{V}^{\text{op}}$ is:

$$\widehat{\alpha}_{P, \pi} : \left\langle \mathbf{y}(F[\mathbf{x}]) \xrightarrow{\beta} P \right\rangle_{j=1}^1 \mapsto \left\langle \mathbf{y}(F[\mathbf{x}]) \xrightarrow{\beta \circ \langle \mathbf{y}\alpha_{\mathbf{x}_i} \rangle_{i=1}^{|\mathbf{x}|}} P \right\rangle$$

To exhibit $\widehat{\quad} : \mathbf{MultiCAT}_{\mathcal{L}}^{\text{coop}} \rightarrow \mathbf{MultiCAT}$ as a co-lax 2-functor, we need two families 2-cells:

- For every locally-small multi-category $\mathcal{U}$, a multi-natural transformation:

$$\widehat{\mathcal{U}} \begin{array}{c} \xrightarrow{\text{Id}_{\widehat{\mathcal{U}}}} \\ \Downarrow \Upsilon_{\mathcal{U}} \\ \xrightarrow{\text{Id}_{\widehat{\mathcal{U}}}} \end{array} \widehat{\mathcal{U}}$$

It is given by factorising the identity multi-natural transformation between the Yoneda embedding and itself:

$$\mathcal{U} \begin{array}{c} \xrightarrow{\mathbf{y}} \widehat{\mathcal{U}} \\ \searrow \text{Id} \\ \xrightarrow{\mathbf{y}} \widehat{\mathcal{U}} \end{array} \quad \widehat{\mathcal{U}}(\mathbf{y}; -) \begin{array}{c} \xrightarrow{\text{Id}} \widehat{\mathcal{U}} \\ \Downarrow \Upsilon \\ \xrightarrow{\text{Id}} \widehat{\mathcal{U}} \end{array}$$

We denote it by Υ because it is given by the Yoneda lemma. We can see that directly by calculation (the factorisation maps id to $(\mathbf{y} \xrightarrow{u} F) \mapsto \Upsilon(\text{Id}u \circ \text{id}) = \Upsilon u$), but also by noting:

- The components $\Upsilon_{F, \mathbf{a}}$ are natural in $\mathbf{a}$ and so we have a natural multi-transformation $\Upsilon_F : \widehat{\mathcal{U}}(\mathbf{y}; F) \rightarrow F$, and these are further multi-natural in F and so we have a multi-natural transformation $\Upsilon : \widehat{\mathcal{U}}(\mathbf{y}; -) \rightarrow \text{Id}$.
- By Yoneda reduction:

$$((\Upsilon \bullet \mathbf{y}) \circ (\text{map } \mathbf{y}))(u) = \Upsilon(\mathbf{y}u) = u$$

and so Υ is the left extension.

- We will also need, for every triple of locally-small multi-categories and composable multi-functors $\mathcal{U} \xleftarrow{F} \mathcal{V} \xleftarrow{G} \mathcal{W}$ a 2-cell:

$$\widehat{\mathcal{U}} \begin{array}{c} \xrightarrow{\widehat{F \circ G}} \widehat{\mathcal{W}} \\ \Downarrow \gamma_{F, G} \\ \xrightarrow{\widehat{G \circ F}} \widehat{\mathcal{W}} \end{array}$$

Street and Walters axiomatise a more general structure with a stronger axiom. For a composable pair of multi-functors $\mathcal{U} \xleftarrow{H} \mathcal{V} \xleftarrow{G} \mathcal{W}$ consisting of a (necessarily admissible) multi-functor G between locally-small multi-categories and admissible functor K , a 2-cell:

$$\mathcal{U} \begin{array}{c} \xrightarrow{\mathcal{U}(H, -)} \widehat{\mathcal{V}} \\ \searrow \theta_{G, H} \\ \xrightarrow{\mathcal{U}(H \circ G, -)} \widehat{\mathcal{W}} \end{array} \quad \begin{array}{c} \widehat{\mathcal{V}} \\ \Downarrow \widehat{G} \\ \widehat{\mathcal{W}} \end{array}$$

This 2-cell specialises to $\gamma_{F, G}$ taking $\mathcal{U} := \widehat{\mathcal{U}}$ and

$$H := \widehat{\mathcal{U}}(-, (\mathbf{y} \circ F) -) : \widehat{\mathcal{U}} \xleftarrow{\mathbf{y}} \mathcal{U} \xleftarrow{F} \mathcal{V}$$

The defining universal property for $\theta_{G, H}$ is that it's the unique factorisation:

$$\begin{array}{ccc}
\mathcal{W} & \xrightarrow{G} & \mathcal{V} \xrightarrow{H} \mathcal{U} \\
& \searrow \scriptstyle y & \nearrow \scriptstyle \text{map } H \downarrow \mathcal{U}(H, -) \\
& & \widehat{\mathcal{V}} \downarrow \widehat{G} \\
& \searrow \scriptstyle \text{map } (\widehat{\mathcal{V}}(-, G-)) & \nearrow \scriptstyle \mathcal{U}(H \circ G, -) \\
& & \widehat{\mathcal{W}}
\end{array}
\quad \leftrightarrow \quad
\begin{array}{ccc}
& \mathcal{U}(H, -) & \\
& \curvearrowright & \\
\mathcal{U} & \xrightarrow{\theta_{G, H}} & \widehat{\mathcal{V}} \\
& \searrow \scriptstyle \mathcal{U}(H \circ G, -) & \downarrow \widehat{G} \\
& & \widehat{\mathcal{W}}
\end{array}$$

It consists, for each $\mathcal{U}$ object a , and $\mathcal{W}^{\text{op}}$ object $\mathbf{z}$, of the functions:

$$\theta_{G, H, a, \langle \mathbf{z} \rangle_{j=1}^1} : \left(HG[\mathbf{z}] \xrightarrow{u} a \right)_{j=1}^1 \mapsto \left\langle \left(y_i \xrightarrow{v_i} G[\mathbf{z}_i] \right)_{i=1}^{|\mathbf{z}|} \mapsto \left(H[y] \xrightarrow{u \circ (Hv_i)_{i=1}^{|\mathbf{z}|}} a \right) \right\rangle_{\mathbf{y} \in \mathcal{V}^{\text{op}}}$$

Axiom 3 states that the 2-cells Υ and θ are 2-isomorphisms, and in particular making $\widehat{\cdot} : \mathbf{MultiCAT}_{\mathcal{L}}^{\text{coop}} \rightarrow \mathbf{MultiCAT}$ a pseudo-functor. Before we prove it, we'll first characterise the multi-natural isomorphisms, and in presheaf multi-categories that will require us to characterise the natural multi-isomorphisms.

► **Lemma 20.** Isomorphism lemma.

■ Let $\mathcal{U}$ be a multi-category, $P, Q \in \widehat{\mathcal{U}}$ presheaves over $\mathcal{U}$, and $\alpha : (P) \rightarrow Q$ a unary natural multi-transformation. Then α is an isomorphism iff all of its components are bijective.

■ A multi-natural transformation $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$ is an isomorphism (w.r.t. 2-cell vertical composition) iff its components $\alpha_a : (Fa) \rightarrow Ga$ are isomorphisms in $\mathcal{V}$.

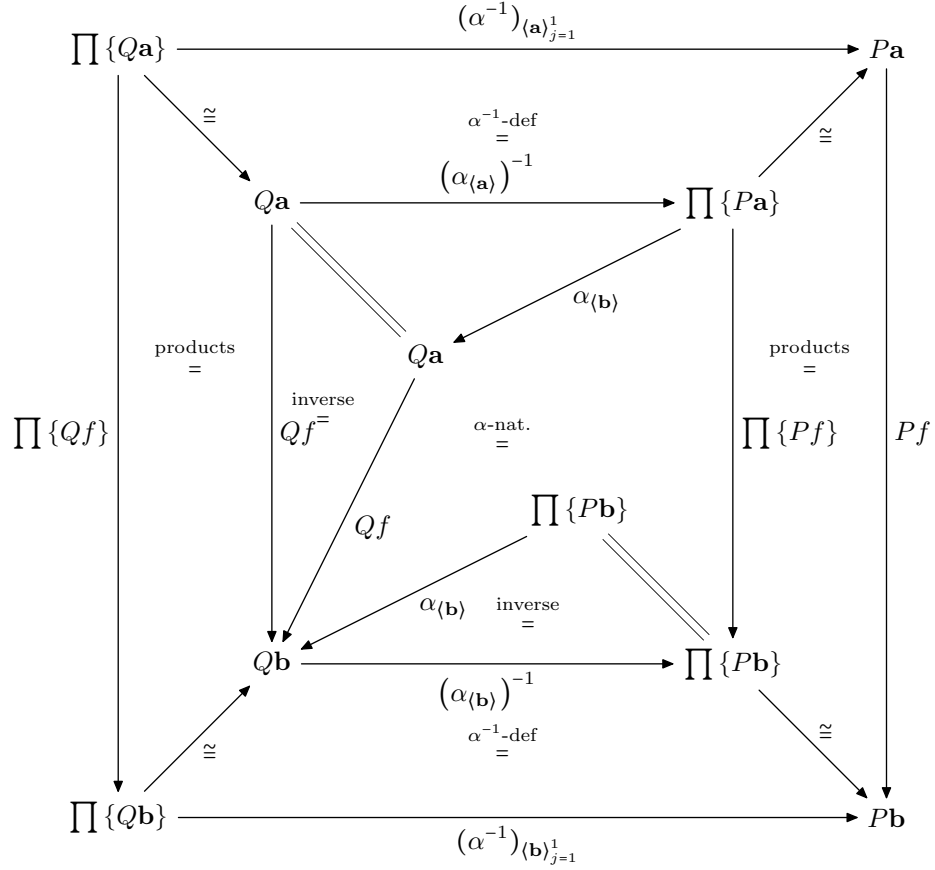
Proof

The ‘only if’ directions follow since the identities are component-wise identity functions/morphisms and the appropriate composition is given component-wise (up-to rearrangement of tuples via a canonical isomorphism). So the remainder of proof amounts to showing that appropriate multi-naturality and naturality conditions for the assembled component-wise inverse.

Starting with a component-wise bijective natural multi-transformation $\alpha : (P) \rightarrow Q$, define its inverse componentwise, at each unary partition $\langle \mathbf{a} \rangle_{j=1}^1$:

$$(\alpha^{-1})_{\langle \mathbf{a} \rangle_{j=1}^1} : \prod \{Q\mathbf{a}\} \xrightarrow{\cong} Q\mathbf{a} \xrightarrow{(\alpha_{\langle \mathbf{a} \rangle_{j=1}^1})^{-1}} \prod \{P\mathbf{a}\} \xrightarrow{\cong} P\mathbf{a}$$

A straightforward calculation shows that this multi-transformation would be the inverse multi-morphism, provided it is indeed natural. Take any $f : \mathbf{a} \leftarrow \mathbf{b}$ in $\mathcal{U}$, and calculate:



Next, consider a multi-natural transformation $\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{G} \end{array} \mathcal{V}$ whose components are multi-isomorphisms. Define $(\alpha^{-1})_a : (Ga) \rightarrow Fa$ by:

$$(\alpha^{-1})_a := (\alpha_a)^{-1}$$

and we now need to show $\alpha^{-1} : G \rightarrow F$ is multi-natural. Take any $f : (a) \rightarrow b$ in $\mathcal{U}$, let $n := |a|$, and calculate:

$$\begin{aligned}
 & Gf \circ \langle (\alpha^{-1})_{a_i} \rangle \xrightarrow{\text{identity} + \alpha^{-1}\text{-def}} \text{id}_b \circ (Gf \circ \langle (\alpha_{a_i})^{-1} \rangle_{i=1}^n) \xrightarrow{\text{inverses}} ((\alpha_b)^{-1} \circ (\alpha_b)) \circ (Gf \circ \langle (\alpha_{a_i})^{-1} \rangle_{i=1}^n) \\
 & \xrightarrow{\text{o-assoc.}} (\alpha_b)^{-1} \circ (\alpha_b \circ (Gf \circ \langle (\alpha_{a_i})^{-1} \rangle_{i=1}^n)) \xrightarrow{\text{multi-naturality}} (\alpha_b)^{-1} \circ (Ff \circ \langle \alpha_{a_i} \circ ((\alpha_{a_i})^{-1}) \rangle_{i=1}^n) \\
 & \xrightarrow{\text{inverses}} (\alpha_b)^{-1} \circ (Ff \circ \langle \text{id}_{a_i} \rangle_{i=1}^n) \\
 & \xrightarrow{\text{identities}} (\alpha_b)^{-1} \circ (Ff)
 \end{aligned}$$

as we want. This proof is the usual one presented algebraically since we don't have a convenient diagrammatic formalism in the multi-categorical setting. $\blacksquare$

► **Proposition 21.** Axiom 3.

- The multi-natural transformation $\Upsilon : \widehat{\text{Id}}_{\mathcal{U}} \rightarrow \text{Id}_{\widehat{\mathcal{U}}}$ is a multi-natural isomorphism.
- The multi-natural transformations $\theta_{G,H} : \mathcal{W}(H \circ G, -) \rightarrow \widehat{G} \circ \mathcal{U}(H, -)$ are multi-natural isomorphisms.

Proof

By the Yoneda lemma, each component of Υ is a bijection, and by the Isomorphism Lemma (1) each component Υ_P is a natural multi-isomorphism, and therefore (2) Υ is a multi-natural isomorphism.

Turning to $\theta_{G,H}$ for $\mathcal{U} \xleftarrow{H} \mathcal{V} \xleftarrow{G} \mathcal{W}$, we will use its UP to show that each component:

$$\theta_{G,H,a,\{\mathbf{z}\}_{k=1}^1} : \mathcal{U}(HG[\mathbf{z}], a) \rightarrow \widehat{\mathcal{V}}(\mathbf{y}[G[\mathbf{z}]], \mathcal{U}(H, a))$$

is in fact the inverse Yoneda Lemma:

$$\Upsilon_{\mathcal{U}, \mathcal{U}(H[-], a), G[\mathbf{z}]}^{-1} : \mathcal{U}(HG[\mathbf{z}], a) \rightarrow \widehat{\mathcal{V}}(\mathbf{y}[G[\mathbf{z}]], \mathcal{U}(H, a))$$

and therefore bijective. Showing the universal property means showing that Υ^{-1} is multi-natural and that:

Multi-naturality follows from the Isomorphism Lemma, and we'll establish the equation by direct calculation, taking any $z \in \text{Ob}(\mathcal{W})$, $\mathbf{z} \in \mathcal{W}^{\text{op}}$ and $w : (\mathbf{z}) \rightarrow z$ in $\mathcal{W}$ and showing that:

$$\begin{aligned} & ((\widehat{G} \bullet (\text{map } H) \bullet G) \circ (\text{map}(\mathbf{y} \circ G)))_{z, \{\mathbf{z}\}_{k=1}^1} w \\ &= ((\Upsilon^{-1} \bullet (G \circ H)) \circ (\text{map}(H \circ G)))_{z, \{\mathbf{z}\}_{k=1}^1} w \end{aligned} \quad (1)$$

Type-checking, this equation is between natural multi-transformations within the multi-hom-set:

$$\widehat{\mathcal{V}}(\mathbf{y}(G[\mathbf{z}]), \mathcal{U}(H, HGz))$$

and so let $n := |\mathbf{z}|$, and take an n -ary partition π of some $\mathbf{y} \in \mathcal{V}^{\text{op}}$, and any n -tuple of multi-morphisms:

$$\left\langle (\pi_i) \xrightarrow{v_i} G\mathbf{z}_i \right\rangle_{i=1}^n \in \prod_{i=1}^n \mathbf{y}(G\mathbf{z}_i)(\pi_i)$$

and apply both sides of (1) at the π -component to this tuple.

On the LHS:

$$\begin{aligned}
& \left(((\widehat{G} \bullet (\text{map } H) \bullet G) \circ (\text{map}(\mathbf{y} \circ G)))_{z, \langle \mathbf{z} \rangle_{k=1}^1} w \right)_{\pi} \langle v_i \rangle_{i=1}^n \\
& \quad \downarrow \begin{array}{c} \widehat{G} \text{ action on natural} \\ \text{multi-transformations} \end{array} \\
& = \left((\widehat{G}(\text{map } H)_{Gz})_{\langle \mathbf{z} \rangle_{k=1}^1} (\mathbf{y}(Gw)) \right)_{\pi} \langle v_i \rangle_{i=1}^n = (\text{map } H \circ (\mathbf{y}(Gw)))_{\pi} \langle v_i \rangle_{i=1}^n \\
& \quad \downarrow \begin{array}{c} \text{map } H\text{-def,} \\ \text{multi-functoriality} \end{array} \\
& = \text{map } H(Gw \circ \langle v_i \rangle_{i=1}^n) = HGw \circ \langle Hv_i \rangle_{i=1}^n
\end{aligned}$$

On the RHS:

$$\begin{aligned}
& \left(((\Upsilon^{-1} \bullet (G \circ H)) \circ (\text{map}(H \circ G)))_{z, \langle \mathbf{z} \rangle_{k=1}^1} w \right)_{\pi} \langle v_i \rangle_{i=1}^n \\
& \quad \downarrow \begin{array}{c} \Upsilon^{-1}\text{-def, map-def} \end{array} \\
& = \Upsilon_{\mathcal{U}(H;a);\mathbf{y}}^{-1} (\text{map}(H \circ G)w) \langle v_i \rangle_{i=1}^n = \mathcal{U}(H[\langle v_i \rangle_{i=1}^n]; a)(HGw) = HGw \circ \langle Hv_i \rangle_{i=1}^n
\end{aligned}$$

and so (1) holds, and $\theta_{G,H}$ is a multi-natural isomorphism. $\blacksquare$

5.8 Representability

We can now reap the fruit of our labour:

► **Proposition 22.** Characterisation: multi-categorical adjunctions. *Let $\mathcal{U}$ be a locally-small multi-category. TFAE:*

- An adjunction $\left(\mathcal{U} \begin{array}{c} \xrightarrow{F} \\ \perp \\ \xleftarrow{U} \end{array} \mathcal{V}, \text{Id} \xrightarrow{\eta} UF, FU \xrightarrow{\varepsilon} \text{Id} \right)$ in **MultiCAT** with F admissible.
- A multi-natural isomorphism $\begin{array}{ccc} & \mathcal{U} & \\ U \nearrow & \varphi \cong & \searrow u(-, -) \\ \mathcal{V} & & \widehat{\mathcal{U}} \\ & \mathcal{V}(F, -) & \end{array}$ with F admissible.
- A multi-functor $U : \mathcal{V} \rightarrow \mathcal{U}$, a choice of objects $Fa \in \mathcal{V}$ for all $a \in \mathcal{U}$ together with a (multi-)morphism $\eta_a : (a) \rightarrow UFa$ such that:
 - All homs $\mathcal{V}(Fa_1, \dots, Fa_n; x)$ are small; and
 - The components of the transformation η are multi-universal: for all $(a_1, \dots, a_n) \xrightarrow{f} Ux$ in $\mathcal{U}$, there exist a unique $(Fa_1, \dots, Fa_n) \xrightarrow{\psi(f)} x$ in $\mathcal{V}$ such that:

$$U(\psi(f)) \circ (\eta_{a_1}, \dots, \eta_{a_n}) = f$$

- An admissible multi-functor $F : \mathcal{U} \rightarrow \mathcal{V}$, a choice of objects $Ux \in \mathcal{U}$ for all $x \in \mathcal{V}$ together with a (multi-)morphism $\varepsilon_x : (FUx) \rightarrow x$ that is universal: for all $(Fa_1, \dots, Fa_n) \xrightarrow{g} x$ in $\mathcal{V}$, there exist a unique $(a_1, \dots, a_n) \xrightarrow{\varphi g} Ux$ in $\mathcal{U}$ such that:

$$\varepsilon_x \circ (F(\varphi g)) = g$$

Proof

The first two items are Prop. 8 of Street and Walters (1978). Given a multi-natural isomorphism φ with inverse ψ , take:

$$\eta_a := \varphi \text{id} \quad \varepsilon_x := \psi \text{id}$$

Take any $f : (\mathbf{a}) \rightarrow Ux$, $n := |\mathbf{a}|$, and chase $\langle \text{id}_{a_n} \rangle_{i=1}^n \in \prod_{i=1}^n \mathcal{V}(Fa_i; Fa_i)$ around the multi-naturality condition for φ when instantiated for ψf in $\mathcal{V}$. OTOH:

$$\varphi(\mathcal{V}(F; \psi f) \langle \text{id}_{a_i} \rangle_{i=1}^n) = \varphi(\psi f \circ \langle \text{id}_{a_i} \rangle_{i=1}^n) = \varphi(\psi f) = f$$

and OTOH:

$$\mathcal{U}(\text{Id}; U(\psi f)) \langle \varphi(\text{id}_{a_i}) \rangle_{i=1}^n = U(\psi f) \circ \langle \eta_{a_i} \rangle_{i=1}^n$$

and we proved the third item.

Take any $g : (F[\mathbf{a}]) \rightarrow x$ in $\mathcal{V}$, and chase id_x around the naturality condition for ψ_x when applied to $\varphi g : \langle Ux \rangle_{k=1}^1 \leftarrow \mathbf{a}$. OTOH:

$$\psi(\mathcal{U}(\varphi g; Ux) \text{id}) = \psi(\text{id} \circ (\varphi g)) = \psi(\varphi g) = g$$

and OTOH:

$$\mathcal{V}(F(\varphi g); Ux)(\psi \text{id}_x) = \varepsilon_x \circ (F\varphi g)$$

as we wanted.

Therefore, the first two conditions imply the last two conditions. We'll now show that the third and fourth conditions imply the third. The exists-unique conditions construct the required bijections, and we need to equip F or U with a multi-functorial action and show that the given bijection is multi-natural in x and natural in $\mathbf{a}$, in each case.

First, assume we have a choice of multi-universal units. Extend F to a functor, by sending each $f : (\mathbf{a}) \rightarrow b$ in $\mathcal{U}$ to:

$$\psi(\eta_b \circ (f)) \quad \text{i.e., the unique } g : (F[\mathbf{a}]) \rightarrow Fb \text{ satisfying:} \quad Ug \circ \langle \eta_{\mathbf{a}_i} \rangle_{i=1}^{|\mathbf{a}|} = \eta_b \circ (f)$$

This assignment is functorial:

$$\begin{array}{ccc} U \text{ multi-functor} & & \\ \downarrow & & \\ U \text{id}_{Fa} \circ (\eta_a) = \eta_a = \eta_a \circ (\text{id}_a) & \implies & F \text{id}_a = \text{id}_{Fa} \end{array}$$

and for all composable $f : (\mathbf{b}) \rightarrow c$, $n := |\mathbf{b}|$ and $\mathbf{f} := \langle f_i : (\mathbf{a}_{(i,*)}) \rightarrow \mathbf{b}_i \rangle_{i=1}^n$, $p := |\mathbf{a}|, m_i := |\mathbf{a}_{(i,*)}|$, we have:

$$\begin{array}{ccc} U \text{ multi-functor} & & \text{o-assoc.} \\ \downarrow & & \downarrow \\ U(Ff \circ \langle F\mathbf{f}_i \rangle_{i=1}^n) \circ \langle \eta_{a_\ell} \rangle_{\ell=1}^p & = & (UFf \circ UF\mathbf{f}_i) \circ \langle \eta_{a_\ell} \rangle_{\ell=1}^p = UFf \circ (UF\mathbf{f}_i \circ \langle \eta_{a_{(i,j)}} \rangle_{j=1}^{m_i}) \\ & = & UFf \circ (\eta_{\mathbf{b}_i} \circ (\mathbf{f}_i)) = (UFf \circ \eta_{\mathbf{b}_i}) \circ (\mathbf{f}) = (\eta_c \circ UFf) \circ (\mathbf{f}) \\ & = & \eta_c \circ (UFf \circ (\mathbf{f})) \end{array}$$

and so $F(f \circ (\mathbf{f})) = Ff \circ (F[\mathbf{f}])$.

Fix x . To show that ψ_x is natural, take any $\mathbf{a} \xleftarrow{\mathbf{u}} \mathbf{b}$ in $\mathcal{U}$, and chasing any $f : (\mathbf{a}) \rightarrow Ux$ around the naturality condition for ψ gives us the proof obligation:

$$\psi(f \circ (\mathbf{u})) = \psi f \circ (F[\mathbf{u}])$$

Calculate:

$$\begin{aligned}
 & \begin{array}{c} F\text{-multi-functor} \\ \circ\text{-assoc.} \\ \downarrow \end{array} U(\psi f \circ (F[\mathbf{u}])) \circ \langle \eta_{\mathbf{a}_\ell} \rangle_{\ell=1}^{|\mathbf{b}|} = U(\psi f) \circ \left\langle U F \mathbf{u}_i \circ \langle \eta_{\mathbf{a}_{(i,j)}} \rangle_{j=1}^{|\mathbf{b}_{(i,*)}|} \right\rangle_{i=1}^{|\mathbf{a}|} \begin{array}{c} F\text{-def} \\ \downarrow \end{array} = U(\psi f) \circ \langle \eta_{\mathbf{b}} \circ (\mathbf{u}_i) \rangle_{i=1}^{|\mathbf{a}|} \\
 & \begin{array}{c} \circ\text{-assoc} \\ \psi\text{-def} \\ \downarrow \end{array} = f \circ (\mathbf{u})
 \end{aligned}$$

and so ψ_x is natural.

To show φ is multi-natural, take any $(\mathbf{x}) \xrightarrow{v} y$ in $\mathcal{V}$, and chase an arbitrary tuple $\langle \mathbf{a}_{(i,*)} \xrightarrow{f_i} U \mathbf{x}_i \rangle_{i=1}^{|\mathbf{x}|}$ in the multi-naturality condition to get the proof obligation:

$$\begin{array}{c} U \text{ multi functor} \\ \circ\text{-assoc.} \\ \psi\text{-def} \\ \downarrow \end{array} \psi(Uv \circ (\mathbf{f})) = v \circ (\psi[\mathbf{f}]) \quad \Longleftarrow \quad U(v \circ (\psi[\mathbf{f}])) \circ \langle \eta_{\mathbf{a}_\ell} \rangle_{\ell=1}^{|\mathbf{a}|} = Uv \circ \langle (\mathbf{f}_i) \rangle_{i=1}^{|\mathbf{x}|}$$

and so ψ is multi-natural. By the Isomorphism Lemma, it is a multi-natural isomorphism, and we proved that the third condition implies the second.

The proof that the final condition implies the second is similar. To extend U to a functor, send $g : (\mathbf{x}) \rightarrow y$ in $\mathcal{V}$ to:

$$\varphi(g \circ \langle \varepsilon_{\mathbf{x}_i} \rangle_{i=1}^{|\mathbf{x}|}) \quad \text{i.e., the unique } f : (U[\mathbf{x}]) \rightarrow Uy \text{ satisfying:} \quad \varepsilon_y \circ (Ff) = f \circ \langle \varepsilon_{\mathbf{x}_i} \rangle_{i=1}^{|\mathbf{x}|}$$

This assignment is functorial:

$$\begin{array}{c} F \text{ multi-functor} \\ \downarrow \end{array} \varepsilon \circ (F \text{id}) = \varepsilon \circ (\text{id}) = \text{id} \circ (\varepsilon) \quad \Longrightarrow \quad U \text{id} = \text{id}$$

and for all composable $(\mathbf{x}) \xrightarrow{g} (\mathbf{y}) \xrightarrow{z} z$:

$$\begin{array}{c} F \text{ multi-functor} \\ \circ \text{ assoc.} \\ \downarrow \end{array} \varepsilon_z \circ (F(Ug \circ (U[\mathbf{g}]))) = (\varepsilon_z \circ F U g) \circ (F U [\mathbf{g}]) = g \circ \langle \varepsilon_{\mathbf{y}_i} \circ (F U [\mathbf{g}_i]) \rangle_i = (g \circ (\mathbf{g})) \circ \langle \varepsilon_{\mathbf{x}_\ell} \rangle_\ell$$

and so $U(g \circ (\mathbf{g})) = U g \circ (U[\mathbf{g}])$.

Fix x . To show that φ_x is natural, take any $\mathbf{a} \xleftarrow{\mathbf{u}} \mathbf{b}$ in $\mathcal{U}$ and chase any $v : (F[\mathbf{a}]) \rightarrow x$ through the multi-naturality condition to get the proof obligation:

$$\begin{array}{c} F \text{ multi-functor} \\ \circ \text{ assoc.} \\ \downarrow \end{array} \varphi(v \circ F[\mathbf{u}]) = (\varphi v) \circ (\mathbf{u}) \quad \Longleftarrow \quad \varepsilon_x \circ F((\varphi v) \circ (\mathbf{u})) = (\varepsilon_x \circ F \varphi v) \circ (F[\mathbf{u}]) \begin{array}{c} \varphi\text{-def} \\ \downarrow \end{array} = v \circ (F[\mathbf{u}])$$

Finally, to show that φ is natural, take any $v : (\mathbf{x}) \rightarrow y$ and chase $\mathbf{f} : \mathbf{a} \rightarrow U[\mathbf{x}]$ in the multi-naturality condition to get the proof obligation:

$$\begin{array}{c} F \text{ multi functor} \\ \circ \text{ assoc.} \\ \downarrow \end{array} \varphi(v \circ (\mathbf{f})) = Uv \circ \varphi[\mathbf{f}] \quad \Longleftarrow \quad \varepsilon_y \circ (F(Uv \circ \varphi[\mathbf{f}])) = (\varepsilon_y \circ F U v) \circ (F \varphi[\mathbf{f}]) \begin{array}{c} U\text{-def} \\ \downarrow \end{array} = v \circ \langle \varepsilon_{\mathbf{x}_i} \circ F \varphi[\mathbf{f}] \rangle_i \\
 \begin{array}{c} \varphi\text{-def} \\ \downarrow \end{array} = v \circ (\mathbf{f})$$

as we wanted. ■

6 Multi-homomorphic extension

By generalities, the forgetful functor $\underline{} : \mathbf{Mod}(\mathcal{T}, \mathbf{Set}) \rightarrow \mathbf{Set}$ always has a left adjoint assigning to each set the free $\mathcal{T}$ -algebra over it. This situation does not extend to multi-categories: we may not have a left adjoint (in **MultiCAT**) to the forgetful multi-functor $\underline{} : \mathbf{Multi}(\mathcal{T}, \mathbf{Set}) \rightarrow \langle \mathbf{Set}, \Pi \rangle$. However, we will have such an adjoint in many cases of interest, and our goal is to prove the following characterisation:

► **Theorem 23.** *Let $\mathcal{T}$ be a theory. The forgetful multi-functor $\underline{} : \mathbf{Multi}(\mathcal{T}, \mathbf{Set}) \rightarrow \langle \mathbf{Set}, \Pi \rangle$ has a left adjoint iff $\mathcal{T}$ is commutative.*

The two directions employ quite different techniques, and so we'll split them into two parts: Proposition 27 and Proposition 29.

In the commutative case, the result follows from, or is already present in, the work of Kock (1971) on commutative monads. Let T be a commutative monad on a finitely complete cartesian-closed category $\mathcal{C}$. We can internalise the T -homomorphisms, i.e., the hom-set $\mathcal{C}^T(A, B)$, as the equaliser:

$$\begin{array}{ccccc} \underline{A} \multimap \underline{B} & \xrightarrow{e} & \underline{B}^{\underline{A}} & \xrightarrow{B^{\underline{A}[-]}} & \underline{B}^{T\underline{A}} \\ & & \searrow T & \nearrow B[-]^{T\underline{A}} & \\ & & (T\underline{B})^{T\underline{A}} & & \end{array}$$

and equip with a T -algebra structure defined by this property:

$$\begin{array}{ccc} T\underline{A} \multimap \underline{B} & \xrightarrow{Te} & T(\underline{B}^{\underline{A}}) \\ \downarrow A \multimap B[-] & & \downarrow \lambda \\ \underline{A} \multimap \underline{B} & \xrightarrow{e} & \underline{B}^{\underline{A}} \\ & & \downarrow B[-]^{\underline{A}} \\ & & (T\underline{B})^{\underline{A}} \end{array} \quad :=$$

where the morphism $\lambda : T(Y^X) \rightarrow (TY)^X$ is given by currying the strength and evaluating $T(Y^X) \times X \xrightarrow{\text{str}} T(Y^X \times X) \xrightarrow{\text{eval}} TY$. This construction extends to a functor $(\multimap) : (\text{op } \mathcal{C}^T) \times \mathcal{C}^T \rightarrow \mathcal{C}^T$, as the unique action that makes $e_{A,B} : \underline{A} \multimap \underline{B} \rightarrow \underline{B}^{\underline{A}}$ a natural transformation. This functor equips the Eilenberg-Moore category with the structure of a *closed category* Eilenberg and Kelly (1966), which we will not recount here.

► **Proposition 24.** *Assume $\mathcal{C}$ is complete. For every T -algebra A , the functor $A \multimap - : \mathcal{C}^T \rightarrow \mathcal{C}^T$ preserves small limits.*

I didn't find this proposition anywhere so we'll prove it here, but I'm sure it's well-known.

Proof

Consider a small diagram $D : I \rightarrow \mathcal{C}^T$ and a limiting cone $\langle L, \ell \rangle$ for it. We need to show that $\langle A \multimap L, A \multimap \ell \rangle$ is limiting for $A \multimap D-$. We'll do it in 2 stages:

- We'll show that post-composing with the carrier preserves the limit: $\langle \underline{A} \multimap \underline{L}, \underline{A} \multimap \underline{\ell} \rangle$ is limiting for $\underline{A} \multimap \underline{D}-$.
- We'll use the fact that $\underline{} : \mathcal{C}^T \rightarrow \mathcal{C}$ creates limits to deduce the result.

The first step follows because we can partially-evaluate limits in any order. Let $D' : I' \rightarrow \mathcal{C}$ be the diagram consisting of equaliser diagrams:

$$\begin{array}{ccc} \underline{Di}^A & \begin{array}{c} \xrightarrow{Di^A[-]} \\ \xrightarrow{Di[-]^{TA} \circ T} \end{array} & \underline{Di}^{TA} \end{array}$$

We can calculate this limit in two ways:

- First take the limit w.r.t. i and, since the exponentials $-^A$ and $-^{TA}$ are right adjoints and so preserve limits, get the diagram:

$$\begin{array}{ccc} \underline{L}^A & \begin{array}{c} \xrightarrow{\underline{L}^A[-]} \\ \xrightarrow{\underline{L}[-]^{TA} \circ T} \end{array} & \underline{L}^{TA} \end{array}$$

and then take its limit, which is $A \multimap L$.

- First take the limit w.r.t. the equaliser diagrams to get the diagram $A \multimap \underline{Di}$.

Therefore $\text{Lim}_{i \in I} A \multimap \underline{Di} = \langle A \multimap L, A \multimap \ell \rangle$.

Since $\underline{-} : \mathcal{C}^T \rightarrow \mathcal{C}$ creates limits, it in particular reflects limits, and so $\langle A \multimap L, A \multimap \ell \rangle$ is limiting. ■

► **Proposition 25.** *The functors $A \multimap - : \mathbf{Mod}(\mathcal{T}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{T}, \mathbf{Set})$ have a left adjoint $- \otimes A : \mathbf{Mod}(\mathcal{T}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{T}, \mathbf{Set})$.*

Proof

This proposition follows by generalities. The category $\mathbf{Mod}(\mathcal{T}, \mathbf{Set})$ is complete and cocomplete, and the functor $A \multimap -$ is continuous and satisfies Freyd's solution-set condition. Therefore, by the adjoint functor theorem, the left adjoint exists. ■

This tensor product generalises the linear-hom tensor from linear algebra and its construction in general is non-effective. It equips $\mathcal{C}^T$ with the structure of a symmetric monoidal closed category.

As any Eilenberg-Moore category for a strong monad over a cartesian-closed category, $\mathcal{C}^T$ has powers, i.e., for every set X and T -algebras Γ, A , we have a representation:

$$\frac{X \longrightarrow \mathcal{C}^T(\Gamma, B) \text{ in } \mathbf{Set}}{\Gamma \longrightarrow B^X \text{ in } \mathcal{C}^T}$$

For $\mathcal{C} = \mathbf{Set}$, B^X is given by the componentwise structure on $\underline{B}^X$.

► **Proposition 26.** *Let T be a commutative monad over $\mathbf{Set}$. Then $B^X \cong (FX \multimap B)$ in $\mathbf{Set}^T$.*

Proof

We construct the bijection using naturality:

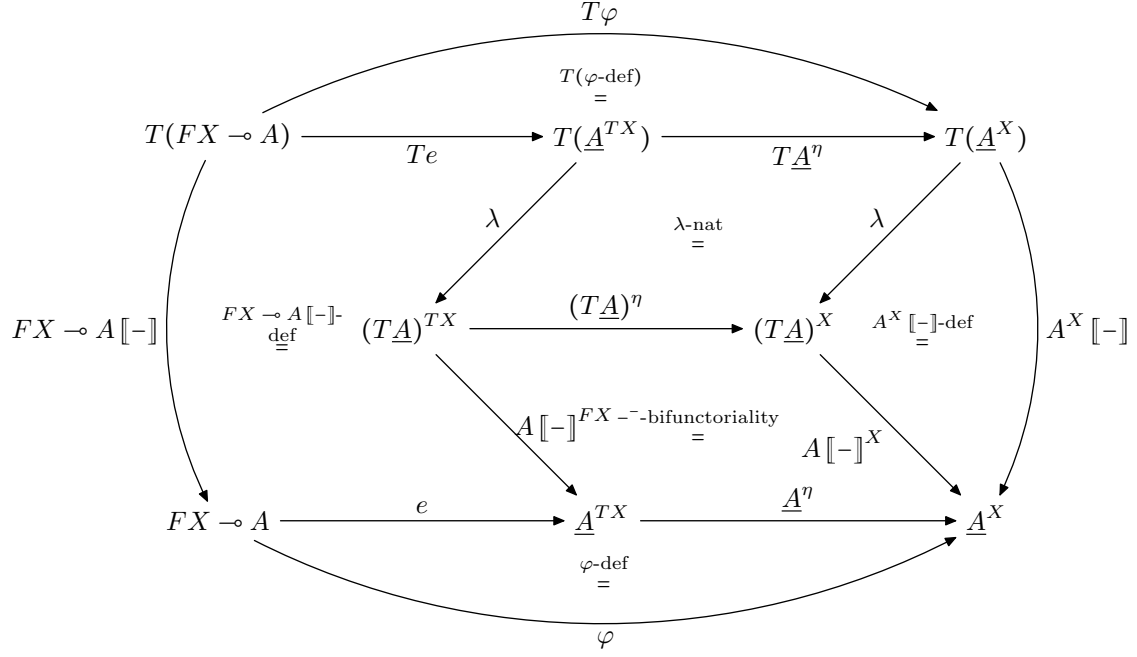
$$\begin{array}{ccc} \underline{1} \longrightarrow \underline{FX \multimap A} & \text{in } \mathbf{Set} \\ \hline \underline{FX} \longrightarrow \underline{A} & \text{in } \mathbf{Set}^T, \text{ by definition} \\ \hline \underline{X} \longrightarrow \underline{A} & \text{in } \mathbf{Set}, \text{ by definition} \\ \hline \underline{1} \longrightarrow \underline{A}^X = \underline{A^X} & \text{in } \mathbf{Set} \end{array}$$

and so we have a bijection of sets $\underline{FX} \multimap A \cong \underline{A}^X$. Chasing elements, this bijection sends $h : \underline{FX} \rightarrow A$ to $h \circ \eta : F \rightarrow \underline{A}$ one way, and $f : X \rightarrow \underline{A}$ to $\llbracket f \rrbracket : \underline{FX} \rightarrow A$ the other way.

We need to show it is a T -isomorphism, and it suffices to show one of these functions to be a T -homomorphism. We'll show that

$$\varphi : \underline{FX} \multimap A \xrightarrow{e} \underline{A}^{TX} \xrightarrow{\underline{A}^{\eta X}} \underline{A}^X$$

is a T -homomorphism:



as we wanted. ■

We can now prove the existence of the adjoint in the commutative case:

► **Proposition 27.** *Let $\mathcal{T}$ be a commutative theory. Then there is a left adjoint to the multifunctor $\llbracket - \rrbracket : \mathbf{Multi}(\mathcal{T}, \langle \mathbf{Set}, \Pi \rangle) \rightarrow \langle \mathbf{Set}, \Pi \rangle$.*

Proof

By Proposition 22, it suffices to show that $\underline{FX}$, the free $\mathcal{T}$ -algebra on X , together with the unit $\eta : X \rightarrow \underline{FX}$ for the left adjoint to the functor $\llbracket - \rrbracket : \mathbf{Mod}(\mathcal{T}, \mathbf{Set}) \rightarrow \mathbf{Set}$, is multi-universal.

To construct the bijection $\psi : \langle \mathbf{Set}, \Pi \rangle(X; \underline{A}) \xrightarrow{\cong} \mathbf{Multi}(\mathcal{T}, \mathbf{Set})(\underline{FX}; A)$, we'll note that:

- The homomorphisms $\bigotimes_{i=1}^{|\mathbf{X}|} F\mathbf{X}_i \rightarrow A$ are precisely the multi-homomorphisms $(\langle F\mathbf{X}_i \rangle_{i=1}^{|\mathbf{X}|}) \rightarrow A$ (taking $\otimes \emptyset := F\mathbf{1}$, the unit for $\otimes$).
- We have a natural bijection:

$$\frac{\bigotimes_{i=1}^{|\mathbf{X}|} F\mathbf{X}_i \longrightarrow A \text{ in } \mathbf{Mod}(\mathcal{T}, \mathbf{Set})}{\prod_{i=1}^{|\mathbf{X}|} \mathbf{X}_i \longrightarrow \underline{A} \text{ in } \mathbf{Set}}$$

and moreover the bijection, moving from top to bottom, is given by $f \mapsto \underline{f} \circ (\eta, \dots, \eta)$.

We prove the second point by induction on $|\mathbf{X}|$. For $|\mathbf{X}| = 0$, we have already proved this fact in Proposition 26. Assume precomposition by the units provides a bijection for $\mathbf{X}$ and consider $\mathbf{X}' := X :: \mathbf{X}$. We then have:

$$\begin{array}{c}
 \frac{FX \otimes \bigotimes_{i=1}^{|\mathbf{X}|} F\mathbf{X}_i \longrightarrow A \quad \text{in } \mathbf{Mod}(\mathcal{T}, \mathbf{Set})}{\bigotimes_{i=1}^{|\mathbf{X}|} F\mathbf{X}_i \longrightarrow FX \multimap A \quad \text{in } \mathbf{Mod}(\mathcal{T}, \mathbf{Set})} \\
 \frac{\bigotimes_{i=1}^{|\mathbf{X}|} F\mathbf{X}_i \longrightarrow A^X \quad \text{in } \mathbf{Mod}(\mathcal{T}, \mathbf{Set}), \text{ by Proposition 26}}{\prod_{i=1}^{|\mathbf{X}|} \mathbf{X}_i \longrightarrow \underline{A}^X = \underline{A}^X \quad \text{in } \mathbf{Set}, \text{ by induction hypothesis}} \\
 \frac{\prod_{i=1}^{|\mathbf{X}|} \mathbf{X}_i \longrightarrow \underline{A}^X \quad \text{in } \mathbf{Set}}{X \times \prod_{i=1}^{|\mathbf{X}|} \mathbf{X}_i \longrightarrow \underline{A}^X \quad \text{in } \mathbf{Set}}
 \end{array}$$

chasing a generic element f down the chain gives:

$$f \mapsto \lambda \mathbf{x} \lambda x. f(x, \mathbf{x}) \mapsto \lambda \mathbf{x} \lambda x. f(\eta x, \mathbf{x}) \mapsto \lambda \mathbf{x} \lambda x. f(\eta x, \prod \eta(\mathbf{x})) \mapsto f \circ (\eta, \langle \eta \rangle_i)$$

as we wanted. ■

We now turn to showing that if there is a left adjoint, then the theory must be commutative. First, we show there is at most one candidate for the left adjoint: the same left adjoint as for homomorphisms:

► **Proposition 28.** *If $_ \dashv : \mathbf{Multi}(\mathcal{T}, \langle \mathbf{Set}, \Pi \rangle) \rightarrow \langle \mathbf{Set}, \Pi \rangle$ has a left adjoint, then it has a left adjoint given by the free $\mathcal{T}$ -algebra and its unit.*

Proof

We use the 2-functor $\langle _ \rangle : \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$ to transport the adjunction, and note that the right adjoint $_ \dashv : \mathbf{Multi}(\mathcal{T}, \langle \mathbf{Set}, \Pi \rangle) \rightarrow \langle \mathbf{Set}, \Pi \rangle$ maps to the usual $_ \dashv : \mathbf{Mod}(\mathcal{T}, \mathbf{Set}) \rightarrow \mathbf{Set}$. Therefore, the object map of the left adjoint is given by the free $\mathcal{T}$ -algebra construction, and the unit is the usual unit. ■

► **Proposition 29.** *If $_ \dashv : \mathbf{Multi}(\mathcal{T}, \langle \mathbf{Set}, \Pi \rangle) \rightarrow \langle \mathbf{Set}, \Pi \rangle$ has a left adjoint, $\mathcal{T}$ is commutative.*

Proof

Take any $t : n$ and $s : m$ in $\mathcal{T}$. We need to show that:

$$t \langle s \langle x_{i,j} \rangle_{i=1}^m \rangle_{i=1}^n = s \langle t \langle x_{i,j} \rangle_{i=1}^n \rangle_{j=1}^m$$

Consider the multi-function $\eta : [n] \times [m] \rightarrow T([n] \times [m])$ and extend it to a multi-homomorphism $f : (F[n], F[m]) \rightarrow F([n] \times [m])$ using the adjunction. Consider the (interpretation of the) terms $t \langle \eta i \rangle_i$ and $s \langle \eta j \rangle_j$ in $F[n]$ and $F[m]$, resp., and calculate:

$$f(t \langle \eta i \rangle_i, s \langle \eta j \rangle_j) = t \langle f(\eta i, s \langle \eta j \rangle_j) \rangle_i = t \langle s \langle f(\eta i, \eta j) \rangle_j \rangle_i = t \langle s \langle \eta(i, j) \rangle_j \rangle_i$$

and similarly:

$$f(t \langle \eta i \rangle_i, s \langle \eta j \rangle_j) = s \langle t \langle \eta(i, j) \rangle_i \rangle_j$$

and so $\mathcal{T}$ is commutative. ■

7 The distributive combination of theories

Let $\mathcal{O}$ be an operad and $\mathcal{T}$ a theory. A *translation* $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{T}$ consists of:

- an assignment, for every $f \in \mathcal{O}_n$, of a term $t \in \mathcal{T}_n$;

such that:

- the identity $\text{id} : \mathcal{O}_1$ is sent to the unique variable projection: $\mathfrak{T}\text{id} = x \in \mathcal{T}_1$; and
- the translation respects composition, for all $f \in \mathcal{O}_n$ composable with $g_i \in \mathcal{O}_{m_i}$, namely:

$$\mathfrak{T}(f \circ (g_1, \dots, g_n)) = \mathfrak{T}f[x_i \mapsto \mathfrak{T}g_i]_{i=1}^n$$

A *renaming* $\rho : n \rightarrow m$ is any function from the set of n variables to the set of m variables. Renamings act on terms covariantly in the evident way. A *surjective* translation $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{T}$ is one where every term $t \in \mathcal{T}_n$ can be expressed as an operator $f \in \mathcal{O}_m$ followed by a renaming $\rho : m \rightarrow n$: $t = \mathfrak{T}f[\rho]$. Let $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{M}$ be a surjective translation. $\mathcal{M}$ is a *$\mathfrak{T}$ -affine* theory if:

- for every operator $g : n$ in $\mathcal{M}$, there is a chosen operation $\tilde{g} \in \mathcal{O}_n$ such that

$$n \vdash_{\mathcal{M}} \mathfrak{T}\tilde{g} = g(x_1, \dots, x_n)$$

- the chosen operations $\tilde{g}$ generate $\mathcal{O}$ under identities and operadic composition
- every axiom of the $\mathcal{M}$ is of the form

$$x_1, \dots, x_n \vdash_{\mathcal{M}} \mathfrak{T}f_1[\rho_1] = \mathfrak{T}f_2[\rho_2]$$

for some $f_1 \in \mathcal{O}_{n_1}$, $f_2 \in \mathcal{O}_{n_2}$ and injective renamings $\rho_1 : n_1 \hookrightarrow n$, $\rho_2 : n_2 \hookrightarrow n$.

► **Proposition 30.** *Let $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{T}$ be a translation, and let $\mathcal{C}$ be a category with chosen finite products $\prod$. There is a carrier-preserving, fully-faithful functor:*

$$\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) : \mathbf{Mod}(\mathcal{T}, \mathcal{C}) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathcal{C}, \prod \rangle)$$

given by:

$$\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) A \llbracket f \rrbracket := A \llbracket \mathfrak{T}f \rrbracket \quad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) h := h$$

Proof

The compatibility with composition and identities of the operad follows from their preservation by $\mathfrak{T}$. Preservation of identities and composition, and fully-faithfulness are evident. ■

Given two signatures Σ_1, Σ_2 , their *coproduct* $\Sigma_1 + \Sigma_2$ is the signature consisting of the disjoint union of operations, but retaining the arities:

$$\Sigma_1 + \Sigma_2 := \Sigma_1 \sqcup \Sigma_2 \quad \text{arity}(\iota_i f) := \text{arity}_{\Sigma_i} f$$

or succinctly:

$$\Sigma_1 + \Sigma_2 := \{\iota_i f : n \mid (f : n) \in \Sigma_i, i = 1, 2\}$$

Given a Σ_i -term-in-context $n \vdash_{\Sigma_i} t$, we denote by $n \vdash_{\Sigma_1 + \Sigma_2} \iota_i[t]$ the $\Sigma_1 + \Sigma_2$ -term-in-context obtained by decorating the operations with ι_i . Similarly, given a Σ_i -equation in context $eq := (n \vdash t_1 = t_2)$ we denote by $\iota_i[eq]$ the equation $n \vdash \iota_i[t_1] = \iota_i[t_2]$.

Let $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$ be a translation from an operad $\mathcal{O}$ to a theory $\mathcal{M}$ and $\mathcal{A}$ be any theory. We define the *distributive combination of $\mathfrak{T}$ over $\mathcal{A}$* to be the theory $\mathfrak{T} \triangleright \mathcal{A}$ given by:

$$\begin{aligned} \Sigma_{\mathfrak{T} \triangleright \mathcal{A}} &:= \Sigma_{\mathcal{M}} + \Sigma_{\mathcal{A}} & \text{Ax}_{\mathfrak{T} \triangleright \mathcal{A}} &:= \iota_1[\text{Ax}_{\mathcal{M}}] \cup \iota_2[\text{Ax}_{\mathcal{A}}] \\ &\cup \left\{ \langle x_i \rangle_{\substack{i=1 \\ i \neq i_0}}^n, \mathbf{y} \vdash \mathfrak{T}f(x_1, \dots, x_{i_0-1}, s\mathbf{y}, x_{i_0+1}, \dots, x_n) \mid \begin{array}{l} (f : n) \in \mathcal{O}_n \\ (s : |\mathbf{y}|) \in \Sigma_{\mathcal{A}} \end{array} \right\} \\ &\quad = s \langle \mathfrak{T}f(x_1, \dots, x_{i_0-1}, \mathbf{y}_j, x_{i_0+1}, \dots, x_n) \rangle_{j=1}^{|\mathbf{y}|} \end{aligned}$$

► **Proposition 31.** Models of the distributive combination (Hyland and Power, 2006). *Let $\mathfrak{T} : \mathcal{O} \rightarrow \mathcal{M}$ be a translation from an operad to a theory, and $\mathcal{A}$ be a theory. The category of models for the distributive combination $\mathfrak{T} \triangleright \mathcal{A}$ is given by the following pullback:*

$$\begin{array}{ccc} \text{Mod}(\mathfrak{T} \triangleright \mathcal{A}, \text{Set}) & \xrightarrow{\neg \mathcal{O} \triangleright \mathcal{A}} & \text{Mod}(\mathcal{O}, \text{Multi}(\mathcal{A}, \text{Set})) \\ \downarrow \neg \mathcal{M} \triangleright & \lrcorner & \downarrow \text{Mod}(\mathcal{O}, \neg) \\ \text{Mod}(\mathcal{M}, \text{Set}) & \xrightarrow{\text{Mod}(\mathfrak{T}, \text{Set})} & \text{Mod}(\mathcal{O}, \langle \text{Set}, \Pi \rangle) \end{array}$$

Proof

A model $A \models \mathfrak{T} \triangleright \mathcal{A}$ consists of:

- an $\mathcal{M}$ -model $\underline{A}_{\mathcal{M} \triangleright}$ interpreting the operations in $\mathcal{M}$
- an $\mathcal{A}$ -model $\underline{A}_{\triangleright \mathcal{A}}$ interpreting the operations in $\mathcal{A}$.

Each of the operation $f \in \mathcal{O}_n$ distributes over the operations in $\mathcal{A}$, and so we have an n -ary multi-homomorphism:

$$\underline{A}_{\mathcal{M} \triangleright} \llbracket \mathfrak{T}f \rrbracket : (\underline{A}_{\triangleright \mathcal{A}}, \dots, \underline{A}_{\triangleright \mathcal{A}}) \rightarrow \underline{A}_{\triangleright \mathcal{A}}$$

in $\text{Multi}(\mathcal{A}, \text{Set})$, and this interpretation is compatible with the operadic structure, i.e., we additionally have:

- an $\mathcal{O}$ -model $\underline{A}_{\mathcal{O} \triangleright \mathcal{A}}$ in $\text{Multi}(\mathcal{A}, \text{Set})$;
- whose underlying $\mathcal{O}$ -model in $\langle \text{Set}, \Pi \rangle$ is $\underline{A}_{\mathcal{O} \rightarrow \mathcal{M}} := \text{Mod}(\mathfrak{T}, \text{Set}) \underline{A}_{\mathcal{M}}$.

I.e., we have an object $\langle \underline{A}_{\mathcal{M} \triangleright}, \underline{A}_{\mathcal{O} \triangleright \mathcal{A}} \rangle$ in the pullback. Conversely, given such an object we can reverse this process and assemble back a corresponding $\mathfrak{T} \triangleright \mathcal{A}$ -model. ■

As Hyland and Power, We can refine this construction into an action $\langle \mathcal{O}, \mathcal{A} \rangle \mapsto \mathcal{O} * \mathcal{A}$ of operads on theories to obtain a universal characterisation of $\mathfrak{T} \triangleright \mathcal{A}$ as a pushout of theories:

$$\begin{array}{ccc} \mathfrak{T} \triangleleft \mathcal{A} & \xleftarrow{\quad} & \mathcal{O} * \mathcal{A} \\ \uparrow & \lrcorner & \uparrow \\ \mathcal{M} & \xleftarrow{\quad} & F\mathcal{O} \end{array}$$

We will not pursue this direction here, but we will use the construction $(*)$.

8 Free algebras

Consider a surjective translation $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{M}$ from an operad to a $\mathfrak{T}$ -affine theory, and let $\mathcal{A}$ be a commutative theory. Let $\mathcal{T} := \mathfrak{T} \triangleright \mathcal{A}$ be the distributive combination of $\mathfrak{T}$ over $\mathcal{A}$. Let X be a set, and we'll now construct $F_{\mathcal{T}}X$, the free $\mathcal{T}$ -algebra over X .

Step 1. Let $A_0 := F_{\mathcal{M}}X \models \mathcal{M}$ be the free $\mathcal{M}$ -algebra over X .

Step 2. Applying the functor:

$$_ \dashv \mathfrak{T} := \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) : \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \rightarrow \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$$

gives an operadic algebra $A_1 := \underline{A_0}_{\mathfrak{T}} \models \mathcal{O}$.

Step 3. Since $\mathcal{A}$ is commutative, by Theorem 23, we have an adjunction $F_{\mathcal{A}} \dashv _ \dashv : \mathbf{Multi}(\mathcal{A}, \mathbf{Set}) \rightarrow \langle \mathbf{Set}, \prod \rangle$ in **MultiCAT**. Applying the 2-functor $\mathbf{Mod}(\mathcal{O}, _) : \mathbf{MultiCAT} \rightarrow \mathbf{CAT}$ to it, we now have a left adjoint $F_{\mathcal{O} \rightarrow \mathcal{A}} := \mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) : \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle) \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$. Take $A_{\mathcal{O} \triangleright \mathcal{A}} := F_{\mathcal{O} \rightarrow \mathcal{A}}A_1 \models \mathcal{O} * \mathcal{A}$.

Step 4. Let $A_{\mathcal{O} \triangleright} := \mathbf{Mod}(\mathcal{O}, _)A_{\mathcal{O} \triangleright \mathcal{A}} \models \mathcal{O}$ be the underlying operadic algebra. For each operation symbol $g : n$ of $\mathcal{M}$, define $A_{\mathcal{M}}[g] := A_{\mathcal{O} \triangleright}[\tilde{g}]$.

► **Lemma 32.** *With the preceding notation:*

- For every $f \in \mathcal{O}_n$, the interpretation $A_{\mathcal{M}}[\mathfrak{T}f]$ is equal to $A_{\mathcal{O} \triangleright}[f]$.
- The resulting algebra validates $\mathcal{M}$, i.e.: $A_{\mathcal{M}} \models \mathcal{M}$.

Proof

Let $X := \underline{A_1}$ and let $\eta_X : X \rightarrow F_{\mathcal{A}}X$ be the unit of the adjunction $F_{\mathcal{A}} \dashv U_{\mathcal{A}}$. By construction of $A_{\mathcal{O} \triangleright \mathcal{A}} = F_{\mathcal{O} \rightarrow \mathcal{A}}A_1$, for every $f \in \mathcal{O}_n$, the operation $A_{\mathcal{O} \rightarrow \mathcal{A}}[f]$ is the unique multi- $\mathcal{A}$ -homomorphism extending

$$X^n \xrightarrow{A_1[f]} X \xrightarrow{\eta_X} F_{\mathcal{A}}X$$

We first prove that, for every $f \in \mathcal{O}_n$, the interpretation in $A_{\mathcal{M}}$ of the term $\mathfrak{T}f$ is equal to $A_{\mathcal{O} \triangleright}[f]$. Since the chosen operations $\tilde{g}$ generate $\mathcal{O}$, we may prove this by induction on the operadic expression of f .

- If $f = \text{id}$, the claim is immediate.
- If $f = \tilde{g}$ is one of the chosen generators, the claim is exactly the definition of $A_{\mathcal{M}}[\tilde{g}]$.
- If $f = h \circ (f_1, \dots, f_k)$ then, using the preservation of operadic composition by $\mathfrak{T}$, the induction hypothesis, and the fact that $A_{\mathcal{O} \triangleright}$ is an $\mathcal{O}$ -model, we get

$$A_{\mathcal{M}}[\mathfrak{T}f] = A_{\mathcal{M}}[\mathfrak{T}h[\mathfrak{T}f_1, \dots, \mathfrak{T}f_k]] = A_{\mathcal{O} \triangleright}[h] \circ (A_{\mathcal{O} \triangleright}[f_1], \dots, A_{\mathcal{O} \triangleright}[f_k]) = A_{\mathcal{O} \triangleright}[f]$$

Now let $x_1, \dots, x_m \vdash_{\mathcal{M}} \mathfrak{T}f_1[\rho_1] = \mathfrak{T}f_2[\rho_2]$ be an axiom of $\mathcal{M}$. For each $i = 1, 2$, define $\Phi_i : X^m \rightarrow F_{\mathcal{A}}X$ by

$$\Phi_i(e) := \eta_X(A_1[f_i](e \circ \rho_i))$$

Since $A_0 \models \mathcal{M}$, these two maps are equal: $\Phi_1 = \Phi_2$. For each $i = 1, 2$, define $\bar{\Phi}_i : F_{\mathcal{A}}X^m \rightarrow F_{\mathcal{A}}X$ by

$$\bar{\Phi}_i(e) := A_{\mathcal{O} \triangleright \mathcal{A}}[f_i](e \circ \rho)$$

Proof

Because ρ_i is injective, each coordinate of e is used in at most one argument of $A_{\mathcal{O} \triangleright \mathcal{A}}$, and therefore $\bar{\Phi}_i$ is a multi- $\mathcal{A}$ -homomorphism in each variable. By construction, it extends Φ_i . By uniqueness of multi- $\mathcal{A}$ -homomorphic extension from generators, we conclude that $\bar{\Phi}_1 = \bar{\Phi}_2$. Hence, for every environment $e : [m] \rightarrow \underline{A_{\mathcal{O} \triangleright \mathcal{A}}}$,

$$A_{\mathcal{O} \triangleright \mathcal{A}} \llbracket f_1 \rrbracket (e \circ \rho_1) = A_{\mathcal{O} \triangleright \mathcal{A}} \llbracket f_2 \rrbracket (e \circ \rho_2)$$

Forgetting from $A_{\mathcal{O} \triangleright \mathcal{A}}$ to $A_{\mathcal{O} \triangleright}$ preserves this equality, so the two sides of the axiom have the same interpretation in $A_{\mathcal{M}}$. Hence, all axioms of $\mathcal{M}$ hold in $A_{\mathcal{M}}$, and therefore $A_{\mathcal{M}} \models \mathcal{M}$. ■

Step 5. Let $A \models \mathfrak{T} \triangleright \mathcal{A}$ be the pullback of $A_{\mathcal{O} \triangleright \mathcal{A}}$ and $A_{\mathcal{M}}$.

► **Lemma 33.** *A is a free $\mathfrak{T} \triangleright \mathcal{A}$ -algebra.*

Proof

We already know that A models $\mathfrak{T} \triangleright \mathcal{A}$. Let's prove that it is free.

Let B by any $\mathfrak{T} \triangleright \mathcal{A}$ -model and $f : X \rightarrow \underline{B}$ be a function. Let $B_{\mathcal{O}} := \mathbf{Mod}(\mathcal{O}, \underline{\quad}) \underline{B_{\mathcal{O} \triangleright \mathcal{A}}}$ and $B_{\mathcal{M}} := \underline{B_{\mathcal{M} \triangleright}}$.

- A_0 is a free $\mathcal{M}$ -algebra over X , therefore we have a unique morphism $h_0 : A_0 \rightarrow B_{\mathcal{M}}$ of $\mathcal{M}$ -algebras such that $h_0 \circ \eta^{A_0} = f$.
- There is also an homomorphism $h_1 : A_1 \rightarrow B_{\mathcal{O}}$ of $\mathcal{O}$ -algebras such that $h_1 \circ \eta_{A_0}^{A_1} = h_0$, and $h_1 = h_0$ since $\eta_{A_0}^{A_1} = \text{id}$.
- B has an $\mathcal{A}$ -structure that distributes over $\mathcal{O}$, therefore thanks to the adjunction $\mathbf{Mod}(\mathcal{O}, F_{\mathcal{A}}) \dashv \mathbf{Mod}(\mathcal{O}, \underline{\quad})$, we have a unique morphism $h_2 : A_{\mathcal{O} \triangleright \mathcal{A}} \rightarrow B$ of $\mathcal{O} * \mathcal{A}$ -algebras such that $h_2 \circ \eta_{A_1}^{A_{\mathcal{O} \triangleright \mathcal{A}}} = h_1$.
- Apply the 2-functor $\mathbf{Mod}(\mathcal{O}, \underline{\quad})$ to the morphism $h_2 : A_{\mathcal{O} \triangleright \mathcal{A}} \rightarrow B$ gives a morphism $h_3 : A_{\mathcal{O} \triangleright} \rightarrow B_{\mathcal{O}}$ of $\mathcal{O}$ -algebras such that $h_3 \circ \eta_{A_{\mathcal{O} \triangleright \mathcal{A}}}^{A_{\mathcal{O} \triangleright}} = h_2$ and $h_3 = h_2$ since $\eta_{A_{\mathcal{O} \triangleright \mathcal{A}}}^{A_{\mathcal{O} \triangleright}} = \text{id}$.
- By the surjectivity of $\mathfrak{T}$ and Lemma 32, we now have a $\mathcal{M}$ -morphism $h_4 = h_3 : A_{\mathcal{M}} \rightarrow B_{\mathcal{M}}$ such that $h_4 \circ \eta_{A_{\mathcal{O} \triangleright}}^{A_{\mathcal{M}}} = h_3$.

Let $t \in \mathcal{M}_n$ and $e : [n] \rightarrow \underline{A_{\mathcal{M}}}$. By surjectivity, there is some $f \in \mathcal{O}_m$ and $\theta : m \rightarrow n$ such that $t = \mathfrak{T}f[\theta]$. We define h_4 as

$$h_4 (A_{\mathcal{M}} \llbracket t \rrbracket e) = h_3 (A_{\mathcal{O} \triangleright} \llbracket f \rrbracket e \circ \theta)$$

This is well-defined by Lemma 32. We then check that h_4 is a morphism of $\mathcal{M}$ -algebras. Since h_3 is a morphism of $\mathcal{O}$ -algebras,

$$\begin{aligned} h_3 (A_{\mathcal{O} \triangleright} \llbracket f \rrbracket e \circ \theta) &= B_{\mathcal{O}} \llbracket f \rrbracket (h_3 \circ e \circ \theta, \dots, h_3 \circ e \circ \theta) \\ &= B_{\mathcal{M}} \llbracket f \rrbracket (h_4 \circ e, \dots, h_4 \circ e) \end{aligned}$$

This proves that $h_4 (A_{\mathcal{M}} \llbracket t \rrbracket e) = B_{\mathcal{M}} \llbracket t \rrbracket (h_4 \circ e)$, so h_4 is a morphism of $\mathcal{M}$ -algebras.

In the end we have the following diagram:

$$\begin{array}{ccccc} X & \xrightarrow{\eta^{A_0}} & \underline{A_0} & \xrightarrow{\text{id}} & \underline{A_1} \\ & \searrow f & \downarrow h_0 & \swarrow h_1 & \downarrow \eta_{A_1}^{A_{\mathcal{O} \triangleright \mathcal{A}}} \\ & & \underline{B} & \xleftarrow{h_2} & \underline{A_{\mathcal{O} \triangleright \mathcal{A}}} \\ & & \uparrow h_4 & \swarrow h_3 & \downarrow \text{id} \\ & & \underline{A_{\mathcal{M}}} & \xleftarrow{\text{id}} & \underline{A_{\mathcal{O} \triangleright}} \end{array}$$

We then consider the walking morphism $I : \{0 \xrightarrow{u} 1\}$ and the two functors $F_1 : I \rightarrow \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ and $F_2 : I \rightarrow \mathbf{Mod}(\mathcal{M}, \mathbf{Set})$ defined by $F_1 0 = A_{\mathcal{O} \triangleright \mathcal{A}}$, $F_1 1 = B_{\mathcal{O} * \mathcal{A}}$, $F_1 u = h_2$ and $F_2 0 = A_{\mathcal{M}}$, $F_2 1 = B_{\mathcal{M}}$, $F_2 u = h_4$. By the construction, we can then use the pullback square to get a unique $H : I \rightarrow \mathbf{Mod}(\mathfrak{T} \triangleright \mathcal{A})$ such that the following diagram commutes:

$$\begin{array}{ccc}
 I & \xrightarrow{F_1} & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
 \downarrow F_2 & \searrow H & \downarrow \mathbf{Mod}(\mathcal{O}, _) \\
 \mathbf{Mod}(\mathfrak{T} \triangleright \mathcal{A}, \mathbf{Set}) & \xrightarrow{_ \circ \triangleright \mathcal{A}} & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \\
 \downarrow _ \mathcal{M} & & \downarrow \mathbf{Mod}(\mathcal{O}, _) \\
 \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) & \xrightarrow{\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})} & \mathbf{Mod}(\mathcal{O}, (\mathbf{Set}, \Pi))
 \end{array}$$

This gives us a unique morphism $h = Hu : A \rightarrow B$ of $\mathfrak{T} \triangleright \mathcal{A}$ -algebras.

We need to show that h extends f along η . $h_2 \circ \eta = h_4 \circ \eta = f$, therefore $h \circ \eta = f$ since η comes from the pullback square. This morphism is unique for a given h_2 and h_4 , but these are also uniquely determined by h_0 and h_1 , which are themselves uniquely determined by f . Therefore h is the unique morphism of $\mathfrak{T} \triangleright \mathcal{A}$ -algebras such that the following diagram commutes:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta} & A \\
 & \searrow f & \downarrow h \\
 & & B
 \end{array}$$

This proves that A is the free $\mathfrak{T} \triangleright \mathcal{A}$ -algebra over X . ■

References

- Samuel Eilenberg and G. Max Kelly. Closed categories. In S. Eilenberg, D. K. Harrison, S. MacLane, and H. Röhl, editors, *Proceedings of the Conference on Categorical Algebra*, pages 421–562, Berlin, Heidelberg, 1966. Springer Berlin Heidelberg. ISBN 978-3-642-99902-4.
- Martin Hyland and John Power. Discrete lawvere theories and computational effects. *Theoretical Computer Science*, 366(1):144–162, 2006. ISSN 0304-3975. <https://doi.org/10.1016/j.tcs.2006.07.007>. URL <https://www.sciencedirect.com/science/article/pii/S0304397506004026>. Algebra and Coalgebra in Computer Science.
- Anders Kock. Closed categories generated by commutative monads. *Journal of the Australian Mathematical Society*, 12(4):405–424, 1971. 10.1017/S1446788700010272.
- Saunders Mac Lane. *Categories for the Working Mathematician*. Graduate Texts in Mathematics. Springer New York, NY, 2nd edition, 1978. ISBN 978-0-387-98403-2. 10.1007/978-1-4757-4721-8.
- Ross Street and Robert Walters. Yoneda structures on 2-categories. *Journal of Algebra*, 50(2):350–379, 1978. ISSN 0021-8693. [https://doi.org/10.1016/0021-8693\(78\)90160-6](https://doi.org/10.1016/0021-8693(78)90160-6). URL <https://www.sciencedirect.com/science/article/pii/0021869378901606>.